

Coordinate Geometry

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Introduction

Cartesian coordinate geometry is one of the tools to solve geometry problems in an algebraic way. The main idea of it is to assign each point in the Euclidean plane a pair of coordinates and then find out the algebraic relationship of the coordinates of the points. It is in particular useful for those who like doing algebraic calculations. Although some may think that using coordinate geometry to deal with geometry problems is just for solving the problems and it disrupts the beauty of geometry, this is exactly the target of most participants of the Olympiad - ‘solving the problems’.

Yet, it should be emphasized that using coordinate geometry to solve a problem is usually more complicated. One of the reasons is that most problems are designed to be solved using pure geometry methods. There is a trend to avoid proposing problems which can be solved by coordinates easily. Therefore, it is strongly recommended to investigate the problem in a more geometric way before applying coordinate geometry, unless it is obviously an easy method or you really hate geometry!

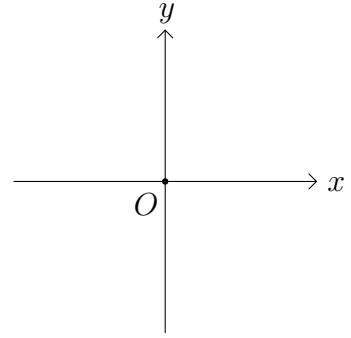
You will soon realize that we need a lot of geometric knowledge even when solving problems with coordinates. With such knowledge, we can find better ways to set up and evaluate the coordinates, so that we can tackle the problem within a reasonable time. Those who are weak in geometry should also find it hard to apply coordinates to solve problems. It might be more appropriate to say that coordinate geometry is an additional tool to help us deduce a proof, rather than a tool that can replace pure geometry.

Sections 1 and 2 introduce the fundamental usages and tricks in coordinate geometry. In sections 3, 4 and 5, we shall see how we can apply coordinates to problems involving angles, circles and incentre/excentres respectively. These are usually considered as objects which could hardly be handled in coordinate geometry.

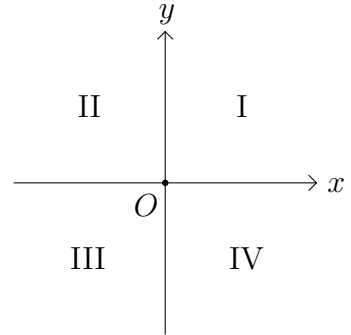
1 Preliminaries

This section is about the most basic knowledge in *coordinate geometry*(座標幾何) and some simple ways of using coordinate geometry to solve geometric problems. We first begin with the setting.

Call one of the points on the plane the *origin*(原點). Usually this point is denoted by O . Two perpendicular lines through O are chosen as the *x-axis*(x -軸) (horizontal) and the *y-axis*(y -軸) (vertical). Then each point P on the plane is assigned a pair of *coordinates*(座標) (a, b) , where a , the *x-coordinate*(x -座標) of P , represents the horizontal distance from O to P , while b , the *y-coordinate*(y -座標) of P , represents the vertical distance from O to P . Here, the term ‘distance’ we use refers to the directed distance. That means the direction is taken into account. For example, the distance 1 is different from -1 .



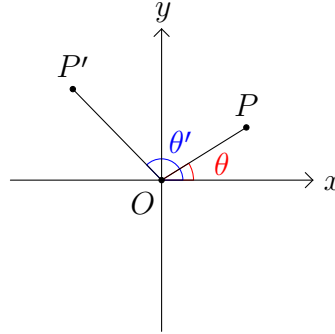
By convention, we set the rightward and upward directions to be the positive x and y directions respectively, as indicated by the arrows on the axes in the figure. The plane with origin O is divided into four parts as shown, usually called *quadrants*(象限) I, II, III, IV. For example, any point (x, y) in quadrant I satisfies $x > 0$ and $y > 0$, while for any point (x', y') in quadrant II, we have $x' < 0$ and $y' > 0$. These set up the foundation of the Cartesian coordinate system. The plane is simply called the *Cartesian plane*(笛卡兒平面) or *coordinate plane*(座標平面).



1.1 Facts in the Cartesian Plane

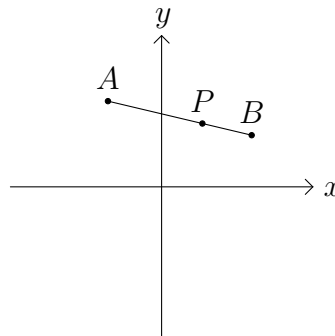
1. By Pythagoras' theorem, we have $OP = \sqrt{a^2 + b^2}$ for any point $P(a, b)$. In general, the distance between any two points $P(a, b)$ and $Q(c, d)$ is given by $PQ = \sqrt{(a - c)^2 + (b - d)^2}$.
2. For any point $P(a, b)$ which is not the origin, let θ be the angle measured anticlockwise from the positive x -axis to the segment OP . This angle is the angle of the polar coordinates of P (this concept is not needed in this note).

Then $\tan \theta = \frac{b}{a}$ (this is indeed the definition of the tangent function). Note that this is true no matter which quadrant P lies in, and it somehow includes the limiting cases when P lies on the y -axis (where both sides are $\pm\infty$).



In general, for distinct points $P(a, b)$ and $Q(c, d)$, the angle θ measured anti-clockwise from the x -axis to the line PQ satisfies $\tan \theta = \frac{d-b}{c-a}$.

3. Let $P(s, t)$ be a point on the segment AB satisfying $AP : PB = \alpha : \beta$, where $A(a, b)$ and $B(c, d)$ are two fixed points. Then we have the sectional formula $s = \frac{\alpha c + \beta a}{\alpha + \beta}$ and $t = \frac{\alpha d + \beta b}{\alpha + \beta}$. This is because the projections of P onto both axes also divide the projections of A, B in the same ratio. In particular, the midpoint of AB has coordinates $\left(\frac{a+c}{2}, \frac{b+d}{2}\right)$. In fact, the formula is valid for every point P on the line AB , where directed lengths are used to express the ratio.

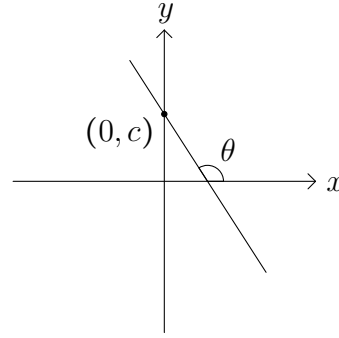


4. Instead of using coordinates, a straight line in the Cartesian plane is represented by an equation in x and y . We shall introduce two important representations here. In particular, we shall mainly use the first representation.

(i) *slope-intercept form*(斜截式): $y = mx + c$

For constants m and c , a point $P(a, b)$ lies on this straight line if and only if (iff) the coordinates satisfy the equation $b = ma + c$.

The value m is called the *slope*(斜率) of the line. Let θ be the angle made between the line and the x -axis (again using our convention for the measurement). It satisfies $m = \tan \theta$. Therefore, two distinct lines are parallel iff they have equal slopes. In this note, we always use the notation m_ℓ to denote the slope of a line ℓ .



The value c is called the *y-intercept*(y-截距) of the line. It means the line cuts the y -axis at the point $(0, c)$.

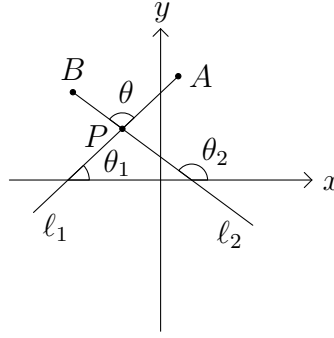
One of the advantages of using this form to represent a straight line is that it is closely related to angles. Also, it is easier to solve equations in this form. However, one should be careful that m could be undefined. This happens when the line is parallel to the y -axis, which has an infinite slope (note that $\tan 90^\circ$ is also undefined). It is certainly better to consider the infinity case separately. But to make things simple, we normally ignore this special case in the proof and assume the same result holds in that case by continuity.

(ii) *general form*(一般式): $Ax + By + C = 0$ where A and B are not both zero

Again, a point $P(a, b)$ lies on the straight line iff $x = a, y = b$ is a solution to this equation. By comparing with the slope-intercept form, we get $m = -\frac{A}{B}$ and $c = -\frac{C}{B}$. The advantage of using the general form is that the equation is still valid for lines parallel to the y -axis, in which case we have $B = 0$.

5. Let θ be the angle measured anticlockwise from a line ℓ_1 to another line ℓ_2 , and let m_1, m_2 be the slopes of ℓ_1, ℓ_2 respectively. Then $\tan \theta = \frac{m_2 - m_1}{1 + m_1 m_2}$ by using the exterior angle of triangle and the compound angle formula for the

tangent function. In particular, we have the relation $\tan \angle APB = \frac{m_{BP} - m_{AP}}{1 + m_{AP}m_{BP}}$ where $\angle APB$ is expressed in directed angles (in other words, $\angle APB$ means the angle measured from PA to PB in the anticlockwise direction). Here, $\tan \theta$ is undefined (when $\ell_1 \perp \ell_2$) iff $m_1 m_2 = -1$.



6. A circle in the coordinate plane is also represented by an equation. There are two usual forms, namely, $(x - a)^2 + (y - b)^2 = r^2$ and $x^2 + y^2 + Dx + Ey + F = 0$. The former one represents a circle with centre (a, b) and radius $r > 0$ (using the distance formula), while the latter one is obtained by expanding the former equation. Both forms are useful in different occasions.
7. Let Γ be a circle with centre $O(a, b)$ and equation $(x - a)^2 + (y - b)^2 = r^2$, and let $P(s, t)$ be a point outside Γ . The length of a tangent from P to Γ is $\sqrt{(s - a)^2 + (t - b)^2 - r^2}$. This is because the length of the tangent is the square root of the power of P with respect to (w.r.t.) Γ . By definition, the power of P is exactly $OP^2 - r^2 = (s - a)^2 + (t - b)^2 - r^2$.
8. The equation of the radical axis of two circles $x^2 + y^2 + D_1x + E_1y + F_1 = 0$ and $x^2 + y^2 + D_2x + E_2y + F_2 = 0$ with different centres is given by the difference of these equations, i.e. $(D_1 - D_2)x + (E_1 - E_2)y + (F_1 - F_2) = 0$. This is because the powers of a point $P(s, t)$ w.r.t. the two circles are $s^2 + t^2 + D_1s + E_1t + F_1$ and $s^2 + t^2 + D_2s + E_2t + F_2$ respectively. Equating these, we obtain the locus of all points having equal powers w.r.t the two circles, and this is exactly

$$(D_1 - D_2)s + (E_1 - E_2)t + (F_1 - F_2) = 0.$$

In particular, if the two circles intersect at two distinct points, this gives the equation of the straight line passing through these two points.

9. (Shoelace formula) Let (x_j, y_j) be the coordinates of A_j for $j = 1, 2, \dots, n$. The area of a convex polygon $A_1A_2 \cdots A_n$ in anticlockwise orientation is

$$\frac{1}{2} \left((x_1y_2 + x_2y_3 + \cdots + x_ny_1) - (x_2y_1 + x_3y_2 + \cdots + x_1y_n) \right).$$

1.2 Basic Usages

As a preliminary section, we first introduce how to use the coordinate plane to solve geometric problems. In this note, we use the abbreviation $\ell_1 \cap \ell_2$ to denote the intersection point of two non-parallel lines ℓ_1 and ℓ_2 .

Problem 1.1. Given points $A(1, -2)$, $B(2, 0)$, $C(1, 6)$, $D(5, -2)$, $E(1, 4)$, $F(5, 0)$, prove that AB, CD, EF are concurrent.

Analysis. After finding out the coordinates of all points in a figure, we need to rephrase the geometric problem to an algebraic one (usually to prove an identity). For example, to prove three lines are concurrent, it means the equations representing these lines have a common root. In particular, note that every pair of equations of straight lines has a unique solution (except when the lines are parallel), which represents the coordinates of the intersection point.

Solution. Note that $m_{AB} = \frac{0 - (-2)}{2 - 1} = 2$. Thus, the equation of AB is of the form $y = 2x + c$. Putting $x = 2$ and $y = 0$, we obtain $c = -4$. This shows the equation of AB is $y = 2x - 4$. Similarly, as $m_{CD} = \frac{6 - (-2)}{1 - 5} = -2$, we find that the equation of CD is $y = -2x + 8$. Solving

$$\begin{cases} y = 2x - 4, \\ y = -2x + 8, \end{cases}$$

we get $(x, y) = (3, 2)$, which are the coordinates of $P = AB \cap CD$. We can find the equation of EF as above, which is $y = -x + 5$. Clearly, $P(3, 2)$ satisfies this equation, and so it lies on EF . This means P, E, F are collinear, and hence AB, CD, EF are concurrent.

Alternatively, we observe that P is the midpoint of EF as $3 = \frac{1+5}{2}$ and $2 = \frac{4+0}{2}$. This also shows P, E, F are collinear (yet this is not a reliable method in general). $\square$

Remarks. In the same way, if we are asked to prove that four points are concyclic, then one straightforward way is to prove that the coordinates of one point satisfy the equation of the circle determined by the other three points.

Problem 1.2. Given points $A(-1, 3)$, $B(-2, -2)$, $C(4, 2)$, find the interior angles of $\triangle ABC$.

Analysis. Sometimes we need to determine the angle or prove that some lines are parallel or perpendicular. One usual way of relating angles in coordinate geometry is to use the slope of the lines. Note that such angles are usually the special angles, whose tangent values are well-known.

Solution. We find the slopes as follows: $m_{AB} = \frac{3 - (-2)}{(-1) - (-2)} = 5$, $m_{BC} = \frac{2 - (-2)}{4 - (-2)} = \frac{2}{3}$ and $m_{CA} = \frac{3 - 2}{(-1) - 4} = -\frac{1}{5}$. Clearly, $m_{AB} \times m_{CA} = -1$ so that AB and AC are perpendicular. This means $\angle A = 90^\circ$.

Next, define $\theta = \angle CBA$. Then

$$\tan \theta = \frac{m_{AB} - m_{BC}}{1 + m_{AB} \cdot m_{BC}} = \frac{5 - \frac{2}{3}}{1 + \frac{10}{3}} = 1.$$

This gives $\angle B = \theta = 45^\circ$ and hence $\angle C = 180^\circ - 90^\circ - 45^\circ = 45^\circ$.

Alternatively, we find

$$\begin{aligned} AB &= \sqrt{((-1) - (-2))^2 + (3 - (-2))^2} = \sqrt{26}, \\ BC &= \sqrt{(4 - (-2))^2 + (2 - (-2))^2} = 2\sqrt{13}, \\ CA &= \sqrt{(4 - (-1))^2 + (2 - 3)^2} = \sqrt{26}. \end{aligned}$$

This also implies $\triangle ABC$ is a right-angled isosceles triangle with $\angle A = 90^\circ$. $\square$

Remarks. One may find $\angle C$ using the same method for checking.

Problem 1.3. (IMO Prelim 2015 Problem 2) Let $ABCD$ be a square of side length 1234. E is a point on CD such that $CEFG$ is a square of side length 567 with F, G outside $ABCD$. The circumcircle of $\triangle ACF$ meets BC again at H . Find CH .

Analysis. It is always easy to set up coordinates when the figure is composed mainly of squares and rectangles. Although this is not the usual case for difficult geometry problems, we demonstrate the basic use of coordinate geometry with this example. We shall also see how circles can be involved.

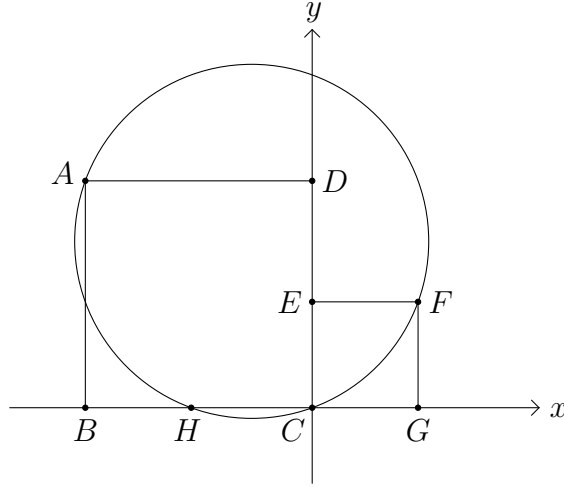
Clearly, the numbers 1234 and 567 do not play any important roles in the solution. It might be simpler to handle the general case.

Solution. We take the common vertex C to be the origin $(0, 0)$. Let $A(-a, a)$, $B(-a, 0)$, $D(0, a)$, $E(0, b)$, $F(b, b)$, $G(b, 0)$ where $a = 1234$ and $b = 567$. We now determine the equation of (ACF) . Let

$$x^2 + y^2 + \alpha x + \beta y + \gamma = 0$$

be the equation of (ACF) . Putting the coordinates of A, C, F respectively, we get

$$\begin{cases} -a\alpha + a\beta + \gamma = -2a^2, \\ \gamma = 0, \\ b\alpha + b\beta + \gamma = -2b^2. \end{cases}$$



Note that α, β, γ are variables to be determined. Using $\gamma = 0$, the other two equations become

$$\begin{cases} -\alpha + \beta = -2a, \\ \alpha + \beta = -2b. \end{cases}$$

Then we easily obtain $\alpha = \frac{-2b - (-2a)}{2} = a - b$ and $\beta = -2b - \alpha = -a - b$. This gives

$$(ACF): \quad x^2 + y^2 + (a - b)x - (a + b)y = 0.$$

To find its intersection with the x -axis, we put $y = 0$. This gives $x^2 + (a - b)x = 0$. The solution $x = 0$ corresponds to the intersection point C , while $x = b - a$ corresponds to H . This shows $H(b - a, 0)$. Then $CH = a - b = 1234 - 567 = 667$. $\square$

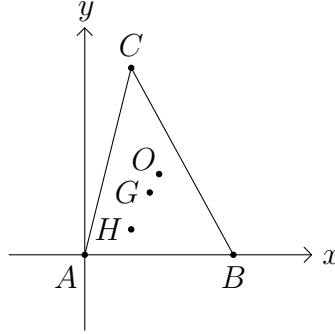
Remarks. In general, it is tough to find the equation of a circle directly, since we need to solve 3 equations in 3 unknowns. A better way is to use some geometric facts to help us or set up the coordinates in a clever way so that the equations can be easily solved. We shall see a number of examples in section 4.

Problem 1.4. (Euler line(歐拉線)) Prove that the circumcentre O , the centroid G and the orthocentre H of $\triangle ABC$ are collinear.

Analysis. Note that the coordinates of G are just the arithmetic means of the coordinates of the three vertices (recall that the centroid is also the centre of mass). To find the coordinates of O and H , it is simpler if one of the sides of $\triangle ABC$ lies horizontally, as perpendicular lines are involved and it is easier to handle equations of lines parallel to the coordinate axes. One simple way is to set A to be the origin and $B(1, 0)$. Note that without loss of generality(WLOG) we may assume $AB = 1$ (or we may regard 1 as 1 unit).

Another common setting is to put B and C on the x -axis, and put A on the y -axis. This setting will be used in other problems later on.

Solution.



Let $A(0,0)$, $B(1,0)$, $C(a,b)$. Clearly, $G = \left(\frac{0+1+a}{3}, \frac{0+0+b}{3} \right) = \left(\frac{1+a}{3}, \frac{b}{3} \right)$.

Next, from $AB \perp CH$, the x -coordinate of H is the same as that of C , which is a . Also, $AC \perp BH$ implies $m_{AC} \cdot m_{BH} = -1$. As $m_{AC} = \frac{b}{a}$, we obtain $m_{BH} = -\frac{a}{b}$. The equation of BH is thus $y = -\frac{a}{b}x + \frac{a}{b}$. Therefore, $H\left(a, \frac{a-a^2}{b}\right)$.

Similarly, since O lies on the perpendicular bisector of AB , its x -coordinate is $\frac{1}{2}$. Also,

$$\begin{aligned} OA &= OC \\ \Rightarrow \left(\frac{1}{2}\right)^2 + y^2 &= \left(a - \frac{1}{2}\right)^2 + (b-y)^2 \\ \Rightarrow 0 &= a^2 - a + b^2 - 2by \\ \Rightarrow y &= \frac{a^2 - a + b^2}{2b}. \end{aligned}$$

This gives $O\left(\frac{1}{2}, \frac{a^2 - a + b^2}{2b}\right)$. The next step is to determine the equation of the Euler line by using any two points among G, H, O . For example, we find

$$m_{GH} = \frac{\frac{b}{3} - \frac{a-a^2}{b}}{\frac{1+a}{3} - a} = \frac{b^2 - 3a + 3a^2}{b(1-2a)},$$

hence the equation of GH is

$$y = \frac{b^2 - 3a + 3a^2}{b(1-2a)}x - \frac{b^2 - 3a + 3a^2}{b(1-2a)} \cdot \frac{1+a}{3} + \frac{b}{3},$$

i.e.

$$y = \frac{b^2 - 3a + 3a^2}{b(1-2a)}x - \frac{a(a^2 - 1 + b^2)}{b(1-2a)}.$$

It remains to check that the coordinates of O satisfy this equation. Indeed,

$$\begin{aligned} \frac{b^2 - 3a + 3a^2}{b(1 - 2a)} \cdot \frac{1}{2} - \frac{a(a^2 - 1 + b^2)}{b(1 - 2a)} &= \frac{-2a^3 + 3a^2 - a + b^2 - 2ab^2}{2b(1 - 2a)} \\ &= \frac{(1 - 2a)(a^2 - a + b^2)}{2b(1 - 2a)} \\ &= \frac{a^2 - a + b^2}{2b}, \end{aligned}$$

so O lies on the line GH . □

Remarks. The last factorization is obvious since the term $1 - 2a$ must be cleared as O does lie on GH .

Alternatively, recall the fact that $OG : GH = 1 : 2$. If we take G' on the segment OH such that $OG' : G'H = 1 : 2$, then

$$G' = \left(\frac{(1)(a) + (2)\left(\frac{1}{2}\right)}{1 + 2}, \frac{(1)\left(\frac{a - a^2}{b}\right) + (2)\left(\frac{a^2 - a + b^2}{2b}\right)}{1 + 2} \right) = \left(\frac{1 + a}{3}, \frac{b}{3} \right).$$

This yields $G = G'$ and hence G lies on the segment OH with ratio $OG : GH = 1 : 2$. This shows knowledge in pure geometry is useful even when working in the Cartesian plane. Note that this fact is also useful to find the coordinates of O , as the coordinates of G and H are easier to find.

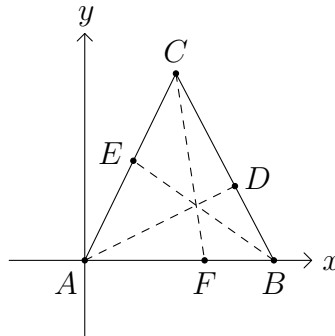
Problem 1.5. (Ceva's theorem(塞瓦定理)) In $\triangle ABC$, let D, E, F be points on the lines BC, CA, AB respectively. Prove that AD, BE, CF are concurrent iff

$$\frac{BD}{DC} \cdot \frac{CE}{EA} \cdot \frac{AF}{FB} = 1,$$

expressed in directed lengths.

Analysis. One of the weaknesses of coordinate geometry is to deal with lengths of segments because it usually involves square roots of the expressions. However, it is easy to express ratios of lengths using the sectional formula. Therefore, the ratios in the given condition can be easily handled.

Solution.



Let $A(0,0)$, $B(1,0)$, $C(a,b)$ and $BD : DC = 1 : k$, $CE : EA = 1 : m$, $AF : FB = 1 : n$.
Then

$$D = \left(\frac{(1)(a) + (k)(1)}{1+k}, \frac{(1)(b) + (k)(0)}{1+k} \right) = \left(\frac{a+k}{1+k}, \frac{b}{1+k} \right)$$

and similarly, $E \left(\frac{ma}{1+m}, \frac{mb}{1+m} \right)$ and $F \left(\frac{1}{1+n}, 0 \right)$. It is easy to find

$$\begin{aligned} AD : \quad y &= \frac{b}{a+k}x, \\ BE : \quad y &= \frac{mb}{ma - (m+1)}x - \frac{mb}{ma - (m+1)}, \\ CF : \quad y &= \frac{b(1+n)}{a(1+n) - 1}x - \frac{b}{a(1+n) - 1}. \end{aligned}$$

Solving the equations of AD and BE , we get $P \left(\frac{m(a+k)}{mk+m+1}, \frac{mb}{mk+m+1} \right)$ where $P = AD \cap BE$. Then

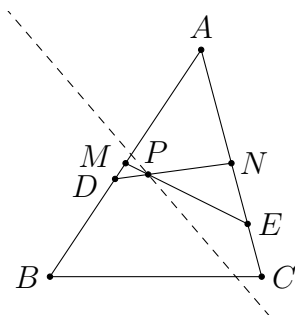
$$\begin{aligned} &AD, BE, CF \text{ are concurrent} \\ \Leftrightarrow &\frac{mb}{mk+m+1} = \frac{b(1+n)}{a(1+n)-1} \cdot \frac{m(a+k)}{mk+m+1} - \frac{b}{a(1+n)-1} \\ \Leftrightarrow &mb(a(1+n)-1) = mb(1+n)(a+k) - b(mk+m+1) \\ \Leftrightarrow &mab(1+n) - mb = mba(1+n) + mbk(1+n) - b(mk+m+1) \\ \Leftrightarrow &-m = mk(1+n) - (mk+m+1) \\ \Leftrightarrow &kmn = 1 \\ \Leftrightarrow &\left(\frac{1}{k} \right) \left(\frac{1}{m} \right) \left(\frac{1}{n} \right) = 1 \\ \Leftrightarrow &\frac{BD}{DC} \cdot \frac{CE}{EA} \cdot \frac{AF}{FB} = 1. \end{aligned}$$

□

2 Symmetry and Algebraic Tricks

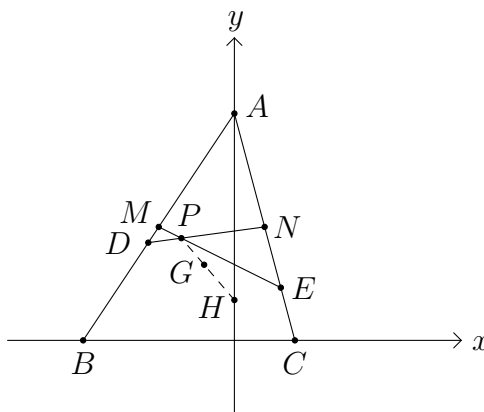
The proofs of the following problems may not be simpler than the geometric proofs. The advantage of using coordinate geometry is that we do not have to think too much about how to find out a solution. We just need to consider how to set up the coordinates so as to minimize the amount of calculations. In this section, we shall introduce some algebraic techniques which help simplify our calculations in many situations. In particular, we shall see that symmetry is a very useful trick.

Problem 2.1. Given $\triangle ABC$, let M and N be the midpoints of AB and AC respectively. Let D be the foot of altitude from C , and E be the foot of altitude from B . Suppose the lines DN and EM meet at P . Prove that P lies on the Euler line of $\triangle ABC$.



Analysis. This time we shall use a standard setting in triangle geometry, i.e. $A(0, a)$, $B(b, 0)$, $C(c, 0)$ (one may even take $a = 1$). This setting has a few advantages. Firstly, the line BC is the x -axis while the A -altitude is the y -axis. Points on these lines or perpendicular lines to them have simple expressions. For example, the orthocentre can be found easily. Secondly, it is useful when the figure has a symmetry between B and C . It means that the construction w.r.t. the two vertices are exactly the same. For instance, points D and E are the corresponding points in this problem.

Solution.



Let $A(0, a)$, $B(b, 0)$, $C(c, 0)$. Then $M\left(\frac{b}{2}, \frac{a}{2}\right)$, $N\left(\frac{c}{2}, \frac{a}{2}\right)$. From $m_{AB} = -\frac{a}{b}$, we find that

$$AB: \quad y = -\frac{a}{b}x + a. \quad (1)$$

As $AB \perp CD$, the slope of CD is $\frac{b}{a}$. Then

$$CD: \quad y = \frac{b}{a}x - \frac{bc}{a}. \quad (2)$$

Solving (1) and (2), we obtain $D\left(\frac{b(a^2 + bc)}{a^2 + b^2}, \frac{ab(b - c)}{a^2 + b^2}\right)$. Now, in order to find the coordinates of E , we simply need to observe the symmetry between D and E . Indeed, by swapping the roles of B and C (meaning to swap the variables b and c), we immediately obtain $E\left(\frac{c(a^2 + bc)}{a^2 + c^2}, \frac{ac(c - b)}{a^2 + c^2}\right)$.

Next, we find the equation of DN . Note that

$$m_{DN} = \frac{\frac{ab(b - c)}{a^2 + b^2} - \frac{a}{2}}{\frac{b(a^2 + bc)}{a^2 + b^2} - \frac{c}{2}} = \frac{2ab(b - c) - a(a^2 + b^2)}{2b(a^2 + bc) - c(a^2 + b^2)} = \frac{a(-a^2 + b^2 - 2bc)}{(2b - c)a^2 + b^2c}.$$

This easily implies

$$DN: \quad y = \frac{a(-a^2 + b^2 - 2bc)}{(2b - c)a^2 + b^2c}x + \frac{ab(a^2 + c^2)}{(2b - c)a^2 + b^2c}. \quad (3)$$

Again by symmetry, we get

$$EM: \quad y = \frac{a(-a^2 + c^2 - 2bc)}{(2c - b)a^2 + bc^2}x + \frac{ac(a^2 + b^2)}{(2c - b)a^2 + bc^2}. \quad (4)$$

Then we solve (3) and (4). This gives

$$\begin{aligned} x &= \frac{\frac{ab(a^2 + c^2)}{(2b - c)a^2 + b^2c} - \frac{ac(a^2 + b^2)}{(2c - b)a^2 + bc^2}}{\frac{a(-a^2 + c^2 - 2bc)}{(2c - b)a^2 + bc^2} - \frac{a(-a^2 + b^2 - 2bc)}{(2b - c)a^2 + b^2c}} \\ &= \frac{b(a^2 + c^2)((2c - b)a^2 + bc^2) - c(a^2 + b^2)((2b - c)a^2 + b^2c)}{(-a^2 + c^2 - 2bc)((2b - c)a^2 + b^2c) - (-a^2 + b^2 - 2bc)((2c - b)a^2 + bc^2)} \\ &= \frac{(c^2 - b^2)a^4 + 2bc(c^2 - b^2)a^2 + b^2c^2(c^2 - b^2)}{3(c - b)a^4 + (b^3 - 9b^2c + 9bc^2 - c^3)a^2 + 3b^2c^2(c - b)}. \end{aligned}$$

Due to symmetry, $c - b$ is usually a factor of the numerator and the denominator. Eliminating this factor, we get

$$x = \frac{(b + c)(a^4 + 2bca^2 + b^2c^2)}{3a^4 - (b^2 - 8bc + c^2)a^2 + 3b^2c^2}.$$

Then

$$\begin{aligned}
y &= \frac{a}{(2b-c)a^2 + b^2c} \cdot ((-a^2 + b^2 - 2bc)x + b(a^2 + c^2)) \\
&= \frac{a((2b-c)a^6 + (5b^2c - 2bc^2)a^4 + (2b^4c - b^3c^2 + 3b^2c^3 - bc^4)a^2 + (b^5c^2 - b^4c^3 + b^3c^4))}{((2b-c)a^2 + b^2c)(3a^4 - (b^2 - 8bc + c^2)a^2 + 3b^2c^2)} \\
&= \frac{a(a^4 + 2bca^2 + (b^3c - b^2c^2 + bc^3))}{3a^4 - (b^2 - 8bc + c^2)a^2 + 3b^2c^2}.
\end{aligned}$$

Note that it is a common feature that the x, y -coordinates of a point always have the same denominator. So it is easy to factorize the numerator to eliminate the term $(2b-c)a^2 + b^2c$. We conclude that

$$P \left(\frac{(b+c)(a^4 + 2bca^2 + b^2c^2)}{3a^4 - (b^2 - 8bc + c^2)a^2 + 3b^2c^2}, \frac{a(a^4 + 2bca^2 + (b^3c - b^2c^2 + bc^3))}{3a^4 - (b^2 - 8bc + c^2)a^2 + 3b^2c^2} \right).$$

It remains to find the equation of the Euler line. Let G and H be the centroid and the orthocentre of $\triangle ABC$ respectively. Note that $G \left(\frac{b+c}{3}, \frac{a}{3} \right)$. From (2), it is obvious that $H \left(0, -\frac{bc}{a} \right)$. We find that

$$GH: \quad y = \frac{a^2 + 3bc}{a(b+c)}x - \frac{bc}{a}. \quad (5)$$

It suffices to show that P satisfies (5). Indeed, it follows from

$$\begin{aligned}
&\frac{a^2 + 3bc}{a(b+c)} \cdot \frac{(b+c)(a^4 + 2bca^2 + b^2c^2)}{3a^4 - (b^2 - 8bc + c^2)a^2 + 3b^2c^2} - \frac{bc}{a} \\
&= \frac{(a^2 + 3bc)(a^4 + 2bca^2 + b^2c^2) - bc(3a^4 - (b^2 - 8bc + c^2)a^2 + 3b^2c^2)}{a(3a^4 - (b^2 - 8bc + c^2)a^2 + 3b^2c^2)} \\
&= \frac{a(a^4 + 2bca^2 + (b^3c - b^2c^2 + bc^3))}{3a^4 - (b^2 - 8bc + c^2)a^2 + 3b^2c^2}.
\end{aligned}$$

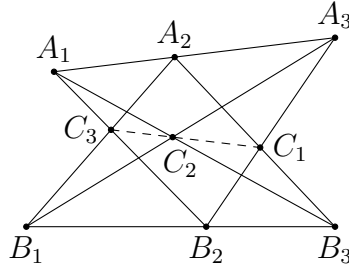
Therefore, P lies on GH , and the conclusion follows. $\square$

Remarks. If $\angle B$ and $\angle C$ are acute, then we may further impose the additional conditions that $b < 0 < c$ and $a > 0$. However, this is usually not needed and not useful in coordinate geometry. In that case, even though b should be negative, we usually do not set the coordinates of B to be $(-b, 0)$ (where $b > 0$ in this case). This is because making this change disrupts the symmetry between the variables b and c , which makes things even worse. One may observe that this symmetry reduces a lot of steps in the above solution.

During the expansion, we write the expressions as polynomials in a . The main reason is still to retain the symmetry between b and c . One may notice that this setting

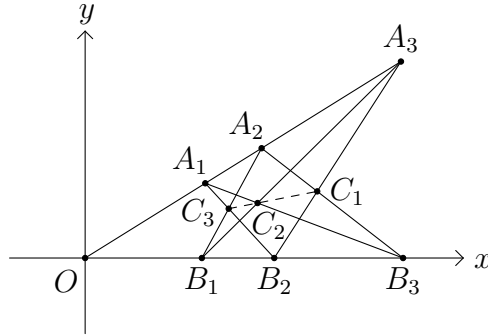
always generates some homogeneous rational functions in a, b, c throughout the steps. We may use this to do some basic checking of the algebraic steps, and this is one advantage of using a instead of some constant like 1.

Problem 2.2. (Pappus' theorem(帕普斯定理)) Let A_1, A_2, A_3 be three points on a line ℓ_1 and let B_1, B_2, B_3 be three points on another line ℓ_2 . Denote the intersection points by $C_3 = A_1B_2 \cap A_2B_1$, $C_2 = A_1B_3 \cap A_3B_1$, $C_1 = A_2B_3 \cap A_3B_2$. Prove that C_1, C_2, C_3 are collinear.



Analysis. One simple way is to take ℓ_2 to be the x -axis. We need not set any B_i to be the origin since it disrupts the symmetry among these points. One way is to take the intersection of ℓ_1 and ℓ_2 to be the origin (if they do intersect) so that the equation of ℓ_1 is simpler. The case when ℓ_1, ℓ_2 are parallel is not considered in the following solution (this may be done separately or be regarded as true in the sense of projective geometry).

Solution.



Set the origin at $O = \ell_1 \cap \ell_2$. Let $B_i(b_i, 0)$, $A_i(a_i, ma_i)$ where $i = 1, 2, 3$ (so that the equation of ℓ_1 is $y = mx$). We first find the coordinates of C_3 . It is easy to deduce

$$A_1B_2: \quad y = \frac{ma_1}{a_1 - b_2}x - \frac{ma_1b_2}{a_1 - b_2}. \quad (1)$$

Swapping the indices 1 and 2, we get

$$A_2B_1: \quad y = \frac{ma_2}{a_2 - b_1}x - \frac{ma_2b_1}{a_2 - b_1}. \quad (2)$$

Solving (1) and (2), we obtain

$$x = \frac{\frac{ma_1b_2}{a_1-b_2} - \frac{ma_2b_1}{a_2-b_1}}{\frac{ma_1}{a_1-b_2} - \frac{ma_2}{a_2-b_1}} = \frac{a_1b_2(a_2-b_1) - a_2b_1(a_1-b_2)}{a_1(a_2-b_1) - a_2(a_1-b_2)} = \frac{a_1a_2b_2 - a_1a_2b_1 - a_1b_1b_2 + a_2b_1b_2}{a_2b_2 - a_1b_1}$$

and

$$\begin{aligned} y &= \frac{ma_1(x-b_2)}{a_1-b_2} \\ &= \frac{ma_1}{a_1-b_2} \cdot \left(\frac{a_1a_2b_2 - a_1a_2b_1 - a_1b_1b_2 + a_2b_1b_2}{a_2b_2 - a_1b_1} - b_2 \right) \\ &= \frac{ma_1}{a_1-b_2} \cdot \frac{a_1a_2b_2 - a_1a_2b_1 + a_2b_1b_2 - a_2b_2^2}{a_2b_2 - a_1b_1} \\ &= \frac{ma_1}{a_1-b_2} \cdot \frac{a_2(a_1-b_2)(b_2-b_1)}{a_2b_2 - a_1b_1} \\ &= \frac{ma_1a_2(b_2-b_1)}{a_2b_2 - a_1b_1}. \end{aligned}$$

Again, the factorization is obvious since the x, y -coordinates should have the same denominator. This gives

$$C_3 \left(\frac{a_1a_2b_2 - a_1a_2b_1 - a_1b_1b_2 + a_2b_1b_2}{a_2b_2 - a_1b_1}, \frac{ma_1a_2(b_2-b_1)}{a_2b_2 - a_1b_1} \right).$$

By changing the indices in the cyclic order $1 \mapsto 2 \mapsto 3 \mapsto 1$, we get

$$C_1 \left(\frac{a_2a_3b_3 - a_2a_3b_2 - a_2b_2b_3 + a_3b_2b_3}{a_3b_3 - a_2b_2}, \frac{ma_2a_3(b_3-b_2)}{a_3b_3 - a_2b_2} \right).$$

Then

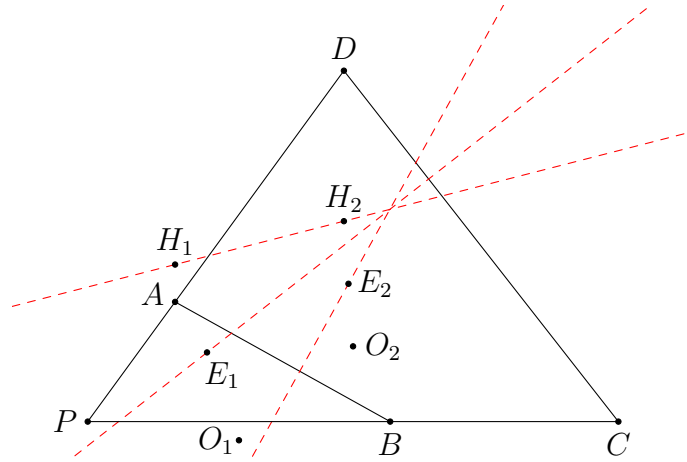
$$\begin{aligned} m_{C_1C_3} &= \frac{\frac{ma_2a_3(b_3-b_2)}{a_3b_3 - a_2b_2} - \frac{ma_1a_2(b_2-b_1)}{a_2b_2 - a_1b_1}}{\frac{a_2a_3b_3 - a_2a_3b_2 - a_2b_2b_3 + a_3b_2b_3}{a_3b_3 - a_2b_2} - \frac{a_1a_2b_2 - a_1a_2b_1 - a_1b_1b_2 + a_2b_1b_2}{a_2b_2 - a_1b_1}} \\ &= \frac{ma_2(a_3(b_3-b_2)(a_2b_2 - a_1b_1) - a_1(b_2-b_1)(a_3b_3 - a_2b_2))}{(a_2a_3b_3 - a_2a_3b_2 - a_2b_2b_3 + a_3b_2b_3)(a_2b_2 - a_1b_1) - (a_1a_2b_2 - a_1a_2b_1 - a_1b_1b_2 + a_2b_1b_2)(a_3b_3 - a_2b_2)} \\ &= \frac{ma_2(a_2a_3b_2b_3 - a_1a_3b_1b_3 - a_2a_3b_2^2 + a_1a_3b_1b_2 - a_1a_3b_2b_3 + a_1a_2b_2^2 + a_1a_3b_1b_3 - a_1a_2b_1b_2)}{(a_2^2a_3b_2b_3 - a_1a_2a_3b_1b_3 - a_2^2a_3b_2^2 + a_1a_2a_3b_1b_2 - a_2^2b_2^2b_3 + a_1a_2b_1b_2b_3 + a_2a_3b_2^2b_3 - a_1a_3b_1b_2b_3) - (a_1a_2a_3b_2b_3 - a_1a_2^2b_2^2 - a_1a_2a_3b_1b_3 + a_1a_2^2b_1b_2 - a_1a_3b_1b_2b_3 + a_1a_2b_1b_2^2 + a_2a_3b_1b_2b_3 - a_2^2b_1b_2^2)} \end{aligned}$$

$$\begin{aligned}
&= \frac{ma_2b_2(a_2a_3b_3 - a_2a_3b_2 + a_1a_3b_1 - a_1a_3b_3 + a_1a_2b_2 - a_1a_2b_1)}{a_2b_2(a_2a_3b_3 - a_2a_3b_2 + a_1a_3b_1 - a_2b_2b_3 + a_1b_1b_3 + a_3b_2b_3 - a_1a_3b_3 + a_1a_2b_2 - a_1a_2b_1 - a_1b_1b_2 - a_3b_1b_3 + a_2b_1b_2)} \\
&= \frac{m(a_1a_2b_2 + a_2a_3b_3 + a_3a_1b_1 - a_1a_2b_1 - a_2a_3b_2 - a_3a_1b_3)}{a_1a_2b_2 + a_2a_3b_3 + a_3a_1b_1 - a_1a_2b_1 - a_2a_3b_2 - a_3a_1b_3 + a_2b_1b_2 + a_3b_2b_3 + a_1b_3b_1 - a_1b_1b_2 - a_2b_2b_3 - a_3b_3b_1}.
\end{aligned}$$

Due to symmetry, C_1C_2 also have the same slope (note that the indices have a cyclic symmetry). Hence, C_1, C_2, C_3 are collinear. $\square$

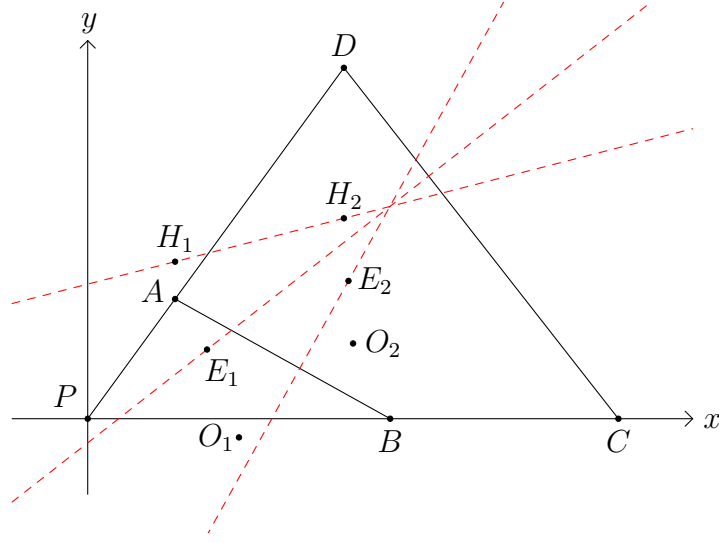
Remarks. This problem highly demonstrates how the symmetry between indices reduces our calculations.

Problem 2.3. (IMO Shortlist 2009 G6) Let the sides AD and BC of the quadrilateral $ABCD$ (such that AB is not parallel to CD) intersect at point P . Points O_1 and O_2 are the circumcentres and points H_1 and H_2 are the orthocentres of triangles ABP and DCP , respectively. Denote the midpoints of segments O_1H_1 and O_2H_2 by E_1 and E_2 , respectively. Prove that the perpendicular from E_1 on CD , the perpendicular from E_2 on AB and the line H_1H_2 are concurrent.



Analysis. This problem seems to involve a lot of points, and the figure is not that easy to draw. If one looks at the figure carefully, one should find that the figure consists of two sets of points which have symmetric constructions, i.e. points w.r.t. $\triangle ABP$ and $\triangle DCP$. To use this symmetry, the simplest way is to set their common point P to be the origin. We may further set PC to be the x -axis so that the perpendicular lines are easy to find.

Solution. Let $P(0,0)$, $B(b,0)$, $C(c,0)$, $A(a,ma)$, $D(d,md)$ where the equation of AD is $y = mx$.



Firstly, we consider $\triangle ABP$. Clearly, the x -coordinate of O_1 is $\frac{b}{2}$. Next, from

$$\begin{aligned}
 & PO_1 = AO_1 \\
 \Rightarrow & \left(\frac{b}{2}\right)^2 + y^2 = \left(a - \frac{b}{2}\right)^2 + (ma - y)^2 \\
 \Rightarrow & 0 = a^2 - ab + m^2a^2 - 2may \\
 \Rightarrow & y = \frac{a - b + m^2a}{2m},
 \end{aligned}$$

we get $O_1\left(\frac{b}{2}, \frac{a - b + m^2a}{2m}\right)$. Next, we have $H_1(a, y)$. To find the y -coordinate, note that

$$\begin{aligned}
 & BH_1 \perp PA \\
 \Rightarrow & m_{BH_1} \cdot m_{PA} = -1 \\
 \Rightarrow & \frac{y}{a - b} \cdot m = -1 \\
 \Rightarrow & y = \frac{b - a}{m}.
 \end{aligned}$$

So $H_1\left(a, \frac{b - a}{m}\right)$ and hence $E_1 = \left(\frac{\frac{b}{2} + a}{2}, \frac{\frac{a - b + m^2a}{2m} + \frac{b - a}{m}}{2}\right) = \left(\frac{b + 2a}{4}, \frac{b - a + m^2a}{4m}\right)$.

By symmetry, we have

$$O_2\left(\frac{c}{2}, \frac{d - c + m^2d}{2m}\right), \quad H_2\left(d, \frac{c - d}{m}\right), \quad E_2\left(\frac{c + 2d}{4}, \frac{c - d + m^2d}{4m}\right).$$

Let ℓ_1 be the perpendicular from E_1 to CD . From $m_{CD} = \frac{md}{d - c}$, we get $m_{\ell_1} = \frac{c - d}{md}$. The equation of ℓ_1 is

$$y = \frac{c - d}{md}x - \frac{(c - d)(b + 2a)}{4md} + \frac{b - a + m^2a}{4m},$$

i.e.

$$\ell_1: 4mdy = 4(c-d)x + (2bd + ad - bc - 2ac + m^2ad). \quad (1)$$

Let ℓ_2 be the perpendicular from E_2 to AB . Swapping the variables $a \leftrightarrow d$ and $b \leftrightarrow c$, we get

$$\ell_2: 4may = 4(b-a)x + (2ac + ad - bc - 2bd + m^2ad). \quad (2)$$

Also, the equation of H_1H_2 is

$$y = \frac{(b-a) - (c-d)}{m(a-d)}x - \frac{bd-ac}{m(a-d)},$$

i.e.

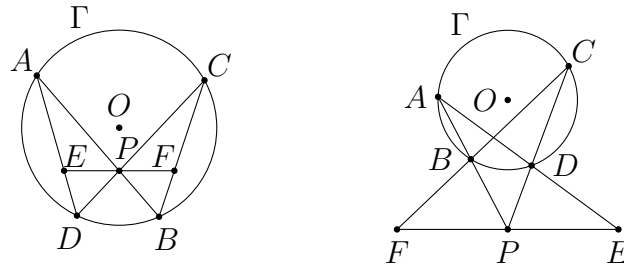
$$H_1H_2: m(a-d)y = ((b-a) - (c-d))x - (bd-ac). \quad (3)$$

We now compare (1), (2) and (3). It is not hard to observe $(2) - (1) = 4 \times (3)$. In other words, as the coordinates of the intersection point $\ell_1 \cap \ell_2$ satisfy both (1) and (2), it also satisfies (3). This means ℓ_1, ℓ_2, H_1H_2 are concurrent. $\square$

Remarks. Since AB is not parallel to CD , we know that ℓ_1 is not parallel to ℓ_2 . Therefore, the case where ℓ_1, ℓ_2, H_1H_2 are parallel is rejected.

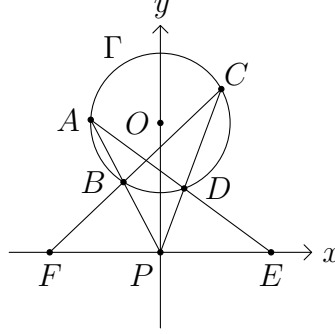
In the above solution, we use some algebraic methods to prove that the three equations have a common root without finding the intersection point explicitly. This is a useful technique, especially when there are many bulky terms. Yet we need a little observation to do so.

Problem 2.4. (Butterfly theorem(蝴蝶定理)) Let Γ be a circle with centre O and let ℓ be a line. Point P lies on ℓ such that $OP \perp \ell$. Two lines passing through P intersect Γ at A, B and C, D respectively. Suppose AD meets ℓ at E and BC meets ℓ at F . Prove that $PE = PF$.



Analysis. Since most of the lines in the figure pass through P , it would be better to set P to be the origin. From the perpendicular condition, we may set O to be on the y -axis so that the equation for the circle is simpler.

Solution.



Let $P(0, 0)$, $O(0, t)$ so that ℓ is the x -axis. Let $x^2 + (y - t)^2 = r^2$ be the equation of Γ . Let $y = mx$ and $y = nx$ be the equations of PA and PC respectively, and let $A(a, ma)$, $B(b, mb)$, $C(c, nc)$, $D(d, nd)$.

Consider the intersection of Γ and AB . Putting $y = mx$ into $x^2 + (y - t)^2 = r^2$, we get $x^2 + (mx - t)^2 = r^2$, which means

$$(m^2 + 1)x^2 - 2mtx + (t^2 - r^2) = 0.$$

Note that a, b are the roots of this quadratic equation. Therefore,

$$a + b = \frac{2mt}{m^2 + 1}, \quad ab = \frac{t^2 - r^2}{m^2 + 1},$$

and similarly,

$$c + d = \frac{2nt}{n^2 + 1}, \quad cd = \frac{t^2 - r^2}{n^2 + 1}.$$

Next, we easily obtain

$$AD: \quad y = \frac{ma - nd}{a - d}x + \frac{ad(n - m)}{a - d}.$$

It cuts the x -axis at $E\left(\frac{ad(m - n)}{ma - nd}, 0\right)$. Due to symmetry, the coordinates of F are $\left(\frac{bc(m - n)}{mb - nc}, 0\right)$.

From these, we find

$$\begin{aligned} & PE = PF \\ \Leftrightarrow & \frac{ad(m - n)}{ma - nd} = -\frac{bc(m - n)}{mb - nc} \\ \Leftrightarrow & ad(mb - nc) + bc(ma - nd) = 0 \\ \Leftrightarrow & mab(c + d) - ncd(a + b) = 0 \\ \Leftrightarrow & m\left(\frac{t^2 - r^2}{m^2 + 1}\right)\left(\frac{2nt}{n^2 + 1}\right) - n\left(\frac{t^2 - r^2}{n^2 + 1}\right)\left(\frac{2mt}{m^2 + 1}\right) = 0. \end{aligned}$$

This is clearly true. □

Remarks. When dealing with equations of circles, we usually need to solve quadratic equations, which probably involve radicals. To avoid this trouble, we can simply use the sum and product of roots instead of finding out the roots explicitly. This is especially useful when handling symmetric expressions.

In this solution, we cover all possible configurations of the butterfly theorem (except that AB or CD is vertical), including the relative positions of ℓ and Γ , and any orders of A, B, C, D . This is a characteristic and advantage of coordinate geometry.

Problem 2.5. Prove the identity

$$(a^2 - b^2)(b^2 - c^2)(c^2 - a^2) = (ab - bc + ca - b^2)(bc - ca + ab - c^2)(ca - ab + bc - a^2).$$

Analysis. This problem is not related to coordinates and geometry. But we always need to prove certain identities when doing coordinate geometry. The important message is that doing factorization should always be the first priority.

Solution. Obviously, $a^2 - b^2 = (a - b)(a + b)$. Also, we have

$$ab - bc + ca - b^2 = b(a - b) + c(a - b) = (a - b)(b + c).$$

Due to symmetry, both sides are equal to $(a - b)(b - c)(c - a)(a + b)(b + c)(c + a)$. $\square$

Remarks. Doing factorizations can simplify the expressions. If this is indeed an identity and one side has a factor $f(a, b, c)$, then the other side must have the same factor. It should be remarked that expanding everything in this problem is a waste of time. This algebraic trick is especially useful in doing coordinate geometry, as can be seen in many examples in later sections.

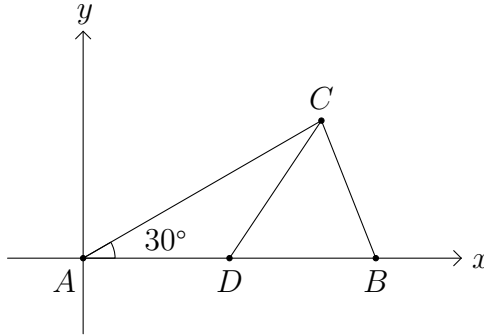
3 Angles

Geometry problems always involve angles. Coordinate geometry does not seem to work well with angles, as the whole system is based on distances. But it is still possible to solve such problems by coordinates. A better way is to do a little angle chasing so as to simplify the conditions involving angles to other conditions like concyclic points, collinear points, etc. If that seems to be impossible, one may apply the relationship between the tangent of angles and the slope of lines to convert everything to algebra.

Problem 3.1. (IMO HK TST 2012 Test 1 Problem 3) In $\triangle ABC$, $\angle A = 30^\circ$ and $AB = 4$. D is the midpoint of AB . Determine the maximum possible value of $\tan \angle DCB$.

Analysis. The only fact needed for this problem is the relation between angles and slope. We demonstrate how to apply this fact to solve this simple problem. As 3 out of the 4 points are collinear, we can set AB to be the x -axis.

Solution.



Set $A(0,0)$, $D(2,0)$, $B(4,0)$. The slope of AC is $\tan 30^\circ = \frac{1}{\sqrt{3}}$ so we can take $C(\sqrt{3}c, c)$ where $c > 0$. Now,

$$m_{CD} = \frac{c-0}{\sqrt{3}c-2} = \frac{c}{\sqrt{3}c-2}, \quad m_{CB} = \frac{c-0}{\sqrt{3}c-4} = \frac{c}{\sqrt{3}c-4}.$$

Then

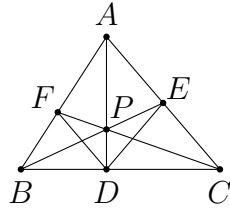
$$\begin{aligned} \tan \angle DCB &= \frac{m_{CB} - m_{CD}}{1 + m_{CB} \cdot m_{CD}} = \frac{\frac{c}{\sqrt{3}c-4} - \frac{c}{\sqrt{3}c-2}}{1 + \frac{c}{\sqrt{3}c-4} \cdot \frac{c}{\sqrt{3}c-2}} = \frac{c((\sqrt{3}c-2) - (\sqrt{3}c-4))}{(\sqrt{3}c-4)(\sqrt{3}c-2) + c^2} \\ &= \frac{2c}{4c^2 - 6\sqrt{3}c + 8} = \frac{c}{2c^2 - 3\sqrt{3}c + 4}. \end{aligned}$$

This is reduced to an inequality problem. To obtain a constant upper bound, we need to turn the denominator to degree 1. The simplest way is to note that $2c^2 + 4 \geq 4\sqrt{2}c$ by the AM-GM inequality. Hence,

$$\tan \angle DCB \leq \frac{c}{(4\sqrt{2} - 3\sqrt{3})c} = \frac{1}{4\sqrt{2} - 3\sqrt{3}} = \frac{4\sqrt{2} + 3\sqrt{3}}{5}.$$

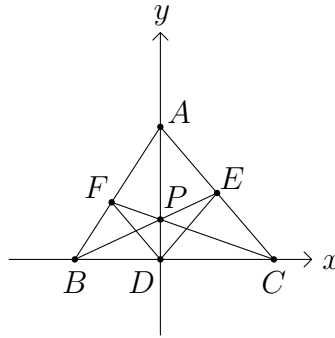
Equality holds when $c = \sqrt{2}$. □

Problem 3.2. (Blanchet's theorem) Let D be the foot of altitude from A of an acute $\triangle ABC$ and let P be a point on the segment AD . Denote by E the intersection of AC and BP , by F the intersection of AB and CP . Prove that $\angle EDP = \angle FDP$.



Analysis. Due to the symmetry between vertices B and C , we shall again apply the standard setting $A(0, a)$, $B(b, 0)$, $C(c, 0)$. For this setting, the altitude AD is the y -axis, hence showing $\angle EDP = \angle FDP$ is the same as showing ED and FD are symmetric in the y -axis, which has a special meaning in terms of the slopes of these lines.

Solution.



Let $A(0, a)$, $B(b, 0)$, $C(c, 0)$, $D(0, 0)$, $P(0, p)$. From $m_{AC} = -\frac{a}{c}$, we find that

$$AC: \quad y = -\frac{a}{c}x + a. \quad (1)$$

Similarly, from $m_{BP} = -\frac{p}{b}$, we get

$$BP: \quad y = -\frac{p}{b}x + p. \quad (2)$$

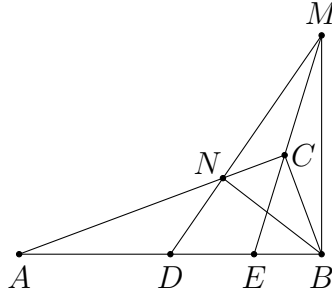
Solving (1) and (2), we obtain $E\left(\frac{bc(a-p)}{ab-cp}, \frac{ap(b-c)}{ab-cp}\right)$. Changing the variables b and c , we have $F\left(\frac{bc(a-p)}{ac-bp}, \frac{ap(c-b)}{ac-bp}\right)$.

We find that $m_{DE} = \frac{ap(b-c)}{bc(a-p)}$ and $m_{DF} = \frac{ap(c-b)}{bc(a-p)}$. Note that $m_{DF} = -m_{DE}$. This means DE and DF are symmetric (the word ‘symmetric’ here really means the existence of a reflection that brings one line to the other) w.r.t. the y -axis (and also the x -axis). Hence, we get $\angle EDP = \angle FDP$. $\square$

Remarks. The condition that $\triangle ABC$ is acute is not necessary. This means the same result that DE and DF are symmetric w.r.t. AD holds equally well in any case (e.g. P lies outside the segment AD). The configuration is needed to deduce $\angle EDP = \angle FDP$ from this fact.

This problem demonstrates one way to show that two lines are symmetric w.r.t. another line ℓ . We may try to take ℓ to be one of the axes so that there is no need to use the angle formula. This setting can also be used to handle an angle bisector condition.

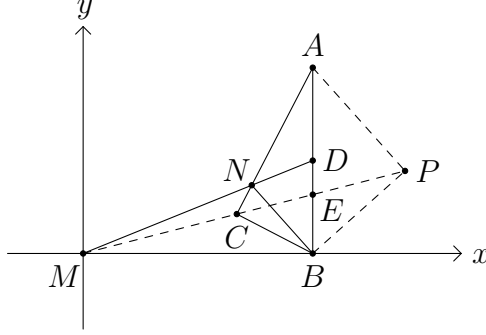
Problem 3.3. (CSMO 2007 Problem 6) In a right-angled triangle ABC , D is the midpoint of the hypotenuse AB , $MB \perp AB$. MD intersects AC at N . The extension of MC intersects AB at E . Prove that $\angle DBN = \angle BCE$.



Analysis. It is usually troublesome to compute the tangent of angles directly. In this problem, we try to move the angles to suitable places so as to simplify the thing to be proved.

Solution. Let P be the point on the line ME such that $BP \perp BN$. Note that

$$\begin{aligned}
 & \angle DBN = \angle BCE \\
 \Leftrightarrow & 90^\circ - \angle DBP = 90^\circ - \angle ACE \\
 \Leftrightarrow & \angle DBP = \angle ACE \\
 \Leftrightarrow & A, C, B, P \text{ are concyclic} \\
 \Leftrightarrow & \angle BPA = 180^\circ - \angle ACB = 90^\circ \\
 \Leftrightarrow & AP \parallel NB.
 \end{aligned}$$



Proving two lines are parallel is usually simpler in coordinate geometry.

Let $M(0,0)$, $B(1,0)$, $E(1,m)$, $C(c,mc)$ so that the equation of ME is $y = mx$. Here, we reconstruct the figure and determine A after fixing M, B, C . This can avoid handling quadratic equations (as C lies on the circle with diameter AB). Such a reconstruction of figure is another important technique in setting the coordinate system.

From $m_{BC} = \frac{mc}{c-1}$, we have $m_{AC} = -\frac{c-1}{mc}$, so that

$$AC: \quad y = -\frac{c-1}{mc}x + \frac{(m^2+1)c-1}{m}. \quad (1)$$

Note that the x -coordinate of A is 1. Then $A\left(1, \frac{(m^2+1)c^2-2c+1}{mc}\right)$ and hence $D\left(1, \frac{(m^2+1)c^2-2c+1}{2mc}\right)$. Thus,

$$MD: \quad y = \frac{(m^2+1)c^2-2c+1}{2mc}x. \quad (2)$$

Solving (1) and (2), we obtain

$$x = \frac{\frac{(m^2+1)c-1}{m}}{\frac{(m^2+1)c^2-2c+1}{2mc} + \frac{c-1}{mc}} = \frac{2c((m^2+1)c-1)}{(m^2+1)c^2-2c+1+2(c-1)} = \frac{2c((m^2+1)c-1)}{(m^2+1)c^2-1}$$

and

$$y = \frac{(m^2+1)c^2-2c+1}{2mc} \cdot \frac{2c((m^2+1)c-1)}{(m^2+1)c^2-1} = \frac{((m^2+1)c^2-2c+1)((m^2+1)c-1)}{m((m^2+1)c^2-1)}.$$

This gives $N\left(\frac{2c((m^2+1)c-1)}{(m^2+1)c^2-1}, \frac{((m^2+1)c^2-2c+1)((m^2+1)c-1)}{m((m^2+1)c^2-1)}\right)$.

Now, we have

$$\begin{aligned}
m_{NB} &= \frac{\frac{((m^2+1)c^2-2c+1)((m^2+1)c-1)}{m((m^2+1)c^2-1)}}{\frac{2c((m^2+1)c-1)}{(m^2+1)c^2-1} - 1} \\
&= \frac{((m^2+1)c^2-2c+1)((m^2+1)c-1)}{m(2c((m^2+1)c-1) - ((m^2+1)c^2-1))} \\
&= \frac{((m^2+1)c^2-2c+1)((m^2+1)c-1)}{m((m^2+1)c^2-2c+1)} \\
&= \frac{(m^2+1)c-1}{m}.
\end{aligned}$$

From $PB \perp NB$, we have $m_{PB} = -\frac{m}{(m^2+1)c-1}$, hence

$$PB: \quad y = -\frac{m}{(m^2+1)c-1}x + \frac{m}{(m^2+1)c-1}.$$

Combining with the equation of ME , which is $y = mx$, we solve the system to get

$$\begin{aligned}
x &= \frac{\frac{m}{(m^2+1)c-1}}{m + \frac{m}{(m^2+1)c-1}} = \frac{1}{(m^2+1)c-1+1} = \frac{1}{(m^2+1)c}, \\
y &= mx = \frac{m}{(m^2+1)c}.
\end{aligned}$$

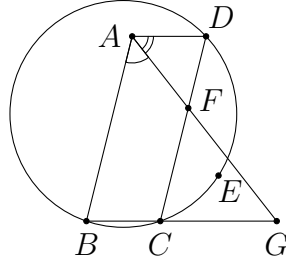
That means $P\left(\frac{1}{(m^2+1)c}, \frac{m}{(m^2+1)c}\right)$. It follows that

$$\begin{aligned}
m_{AP} &= \frac{\frac{(m^2+1)c^2-2c+1}{mc} - \frac{m}{(m^2+1)c}}{1 - \frac{1}{(m^2+1)c}} = \frac{((m^2+1)c^2-2c+1)(m^2+1) - m^2}{m((m^2+1)c-1)} \\
&= \frac{(m^2+1)^2c^2 - 2(m^2+1)c + 1}{m((m^2+1)c-1)} = \frac{((m^2+1)c-1)^2}{m((m^2+1)c-1)} = \frac{(m^2+1)c-1}{m} = m_{NB}.
\end{aligned}$$

This completes the proof. $\square$

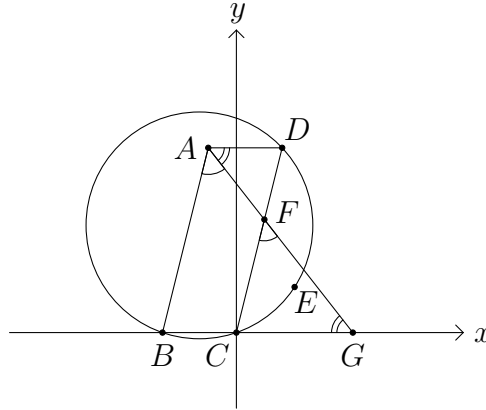
Remarks. Knowledge in pure geometry can often help in changing the conditions and the desired results to more convenient expressions. It is certainly the best if one can solve the problem completely by pure geometry.

Problem 3.4. (IMO Shortlist 2007 G4/Problem 2) Consider five points A, B, C, D and E such that $ABCD$ is a parallelogram and $BCED$ is a cyclic quadrilateral. Let ℓ be a line passing through A . Suppose that ℓ intersects the interior of the segment DC at F and intersects line BC at G . Suppose also that $EF = EG = EC$. Prove that ℓ is the bisector of angle DAB .



Analysis. The figure is composed of a parallelogram $ABCD$ and $\triangle CFG$. Since only point C belongs to both polygons, it is better to set it as the origin. Instead of proving equal angles, we shall prove $\triangle CFG$ is isosceles. Note that we choose not to prove $\triangle BAG$ is isosceles as $\triangle CFG$ involves the origin C while $\triangle BAG$ does not.

Solution.



We first note that

$$\begin{aligned} \angle BAG &= \angle GAD \\ \Leftrightarrow \angle CFG &= \angle FGC \\ \Leftrightarrow CF &= CG \end{aligned}$$

from the parallel lines. Let $C(0,0)$, $B(-1,0)$, $G(g,0)$ (with $g > 0$), $D(d,md)$. As the diagonals of a parallelogram bisect each other, the midpoint of AC coincides with the midpoint of BD . So $A = B + D - C = (d-1, md)$. One way to find F is to solve the equations of AG and CD . Alternatively, we note that $\triangle CFG$ and $\triangle DFA$ are similar. Then

$$\frac{CF}{FD} = \frac{CG}{AD} = g,$$

hence the coordinates of F is $\left(\frac{dg}{g+1}, \frac{mdg}{g+1}\right)$. The x -coordinate of E is $\frac{g}{2}$. For its y -coordinate, note that

$$\begin{aligned}
CE &= EF \\
\Rightarrow \left(\frac{g}{2}\right)^2 + y^2 &= \left(\frac{g}{2} - \frac{dg}{g+1}\right)^2 + \left(y - \frac{mdg}{g+1}\right)^2 \\
\Rightarrow 0 &= \frac{d^2g^2}{(g+1)^2} - \frac{dg^2}{g+1} + \frac{m^2d^2g^2}{(g+1)^2} - \frac{2mdg}{g+1}y \\
\Rightarrow y &= \frac{g(m^2d + d - g - 1)}{2m(g+1)}.
\end{aligned}$$

This gives $E\left(\frac{g}{2}, \frac{g(m^2d + d - g - 1)}{2m(g+1)}\right)$. Similarly, let O be the circumcentre of $\triangle BCD$. Its x -coordinate is $-\frac{1}{2}$. From

$$\begin{aligned}
CO &= DO \\
\Rightarrow \left(-\frac{1}{2}\right)^2 + y^2 &= \left(d + \frac{1}{2}\right)^2 + (y - md)^2 \\
\Rightarrow 0 &= d^2 + d + m^2d^2 - 2mdy \\
\Rightarrow y &= \frac{m^2d + d + 1}{2m},
\end{aligned}$$

the circumcircle of $\triangle BCD$ has equation

$$x^2 + y^2 + x - \frac{m^2d + d + 1}{m}y = 0.$$

It is given that E lies on this circle. We put $x = \frac{g}{2}$ and $y = \frac{g(m^2d + d - g - 1)}{2m(g+1)}$. Note that

$$\begin{aligned}
x(x+1) &= y\left(\frac{m^2d + d + 1}{m} - y\right) \\
\Rightarrow \frac{g}{2}\left(\frac{g}{2} + 1\right) &= \frac{g(m^2d + d - g - 1)}{2m(g+1)} \cdot \left(\frac{m^2d + d + 1}{m} - \frac{g(m^2d + d - g - 1)}{2m(g+1)}\right) \\
\Rightarrow \frac{g+2}{2} &= \frac{m^2d + d - g - 1}{m(g+1)} \cdot \frac{m^2d(g+2) + (g+2)(d+g+1)}{2m(g+1)} \\
\Rightarrow m^2(g+1)^2 &= (m^2d + d)^2 - (g+1)^2 \\
\Rightarrow (m^2 + 1)(g+1)^2 &= (m^2 + 1)d^2 \\
\Rightarrow (g+1)^2 &= (m^2 + 1)d^2. \tag{1}
\end{aligned}$$

(We always try to factorize the expressions whenever possible.)

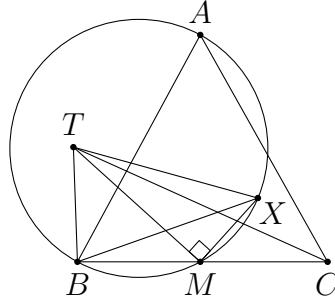
Next, as

$$\begin{aligned}
& CF = CG \\
\Leftrightarrow & \left(\frac{dg}{g+1} \right)^2 + \left(\frac{mdg}{g+1} \right)^2 = g^2 \\
\Leftrightarrow & d^2(m^2 + 1) = (g+1)^2,
\end{aligned}$$

the result follows readily since this is exactly the same as the condition (1). $\square$

Remarks. Doing factorization can always simplify the expressions. If one cannot spot the factorization immediately, one may compare the constraint and the desired equation to obtain the result.

Problem 3.5. (IMO Shortlist 2007 G2) Given an isosceles triangle ABC with $AB = AC$, the midpoint of side BC is denoted by M . Let X be a variable point on the shorter arc MA of the circumcircle of triangle ABM . Let T be the point in the angle domain BMA , for which $\angle TMX = 90^\circ$ and $TX = BX$. Prove that $\angle MTB - \angle CTM$ does not depend on X .



Analysis. There is no symmetry in this problem. Since $\triangle ABC$ is isosceles, it is natural to put M as the origin. The point $X(b, c)$ is a variable point on a circle, which means there is a quadratic polynomial relation between b and c . However, there is no need to express c in terms of b since this certainly involves radicals.

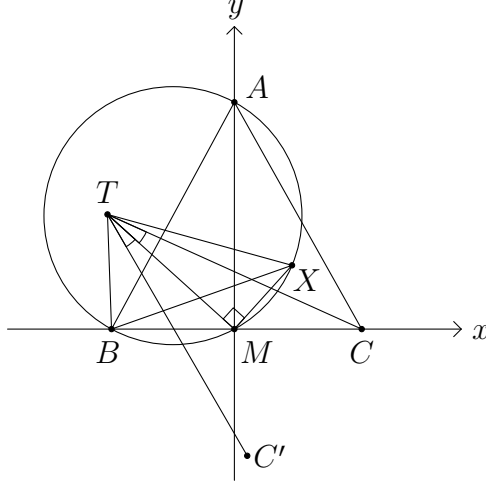
One way to prove the assertion is to compute $\tan(\angle MTB - \angle CTM)$ directly. Another way is to subtract the angles geometrically so that we only need to compute the tangent of a single angle.

Solution. Let $B(-2, 0)$, $M(0, 0)$, $C(2, 0)$, $A(0, 2a)$ and $X(b, c)$ where $b, c > 0$. The centre of (ABM) is the midpoint of AB , which is $(-1, a)$. Hence, the equation of (ABM) is $x^2 + y^2 + 2x - 2ay = 0$. It follows that

$$b^2 + c^2 + 2b - 2ac = 0. \quad (1)$$

Next, $m_{MX} = \frac{c}{b}$ implies $m_{MT} = -\frac{b}{c}$, so that

$$MT: \quad y = -\frac{b}{c}x. \quad (2)$$



Also, T lies on the circle with centre X passing through B , which has equation

$$x^2 + y^2 - 2bx - 2cy - 4b - 4 = 0. \quad (3)$$

Combining (2) and (3), we get

$$\left(\frac{b^2}{c^2} + 1\right)x^2 = 4b + 4.$$

From the relative position, T lies in the second quadrant so that we should choose the negative root for x . We find $T\left(-2c\sqrt{\frac{b+1}{b^2+c^2}}, 2b\sqrt{\frac{b+1}{b^2+c^2}}\right)$.

Instead of finding the tangent of angles directly, we reflect C in the line MT to point C' so that $\angle MTB - \angle CTM = \angle BTC'$. We have $m_{CC'} = \frac{c}{b}$ and hence

$$CC' : y = \frac{c}{b}x - \frac{2c}{b}. \quad (4)$$

The midpoint of CC' is the intersection of MT and CC' . Solving (2) and (4), its coordinates are $\left(\frac{2c^2}{b^2+c^2}, -\frac{2bc}{b^2+c^2}\right)$. Hence, we get

$$C' = 2\left(\frac{2c^2}{b^2+c^2}, -\frac{2bc}{b^2+c^2}\right) - (2, 0) = \left(\frac{2(c^2-b^2)}{b^2+c^2}, -\frac{4bc}{b^2+c^2}\right).$$

To compute $\tan \angle BTC'$, we first find

$$m_{TC'} = \frac{2b\sqrt{\frac{b+1}{b^2+c^2}} + \frac{4bc}{b^2+c^2}}{-2c\sqrt{\frac{b+1}{b^2+c^2}} - \frac{2(c^2-b^2)}{b^2+c^2}} = \frac{b\sqrt{(b+1)(b^2+c^2)} + 2bc}{-c\sqrt{(b+1)(b^2+c^2)} + b^2 - c^2}$$

and

$$m_{TB} = \frac{2b\sqrt{\frac{b+1}{b^2+c^2}}}{-2c\sqrt{\frac{b+1}{b^2+c^2}}+2} = \frac{b\sqrt{b+1}}{-c\sqrt{b+1}+\sqrt{b^2+c^2}}.$$

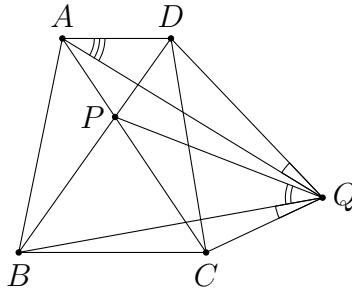
Then

$$\begin{aligned} \tan \angle BTC' &= \frac{m_{TC'} - m_{TB}}{1 + m_{TC'} \cdot m_{TB}} \\ &= \frac{-bc(b+1)\sqrt{b^2+c^2} - 2bc^2\sqrt{b+1} + b(b^2+c^2)\sqrt{b+1} + 2bc\sqrt{b^2+c^2} + bc(b+1)\sqrt{b^2+c^2} - b(b^2-c^2)\sqrt{b+1}}{c^2(b+1)\sqrt{b^2+c^2} - c(b^2-c^2)\sqrt{b+1} - c(b^2+c^2)\sqrt{b+1} + (b^2-c^2)\sqrt{b^2+c^2} + b^2(b+1)\sqrt{b^2+c^2} + 2b^2c\sqrt{b+1}} \\ &= \frac{2bc\sqrt{b^2+c^2}}{b(b^2+c^2+2b)\sqrt{b^2+c^2}} \\ &= \frac{2c}{b^2+c^2+2b} \\ &= \frac{1}{a}, \end{aligned}$$

where the last equality arises from (1). Clearly, this depends only on $\triangle ABC$ and is independent of X . $\square$

Remarks. Sometimes if the coordinates involve radicals, there is no need to do rationalization. We simply need to treat the square root sign as another variable. In most cases, they will cancel out each other in the end.

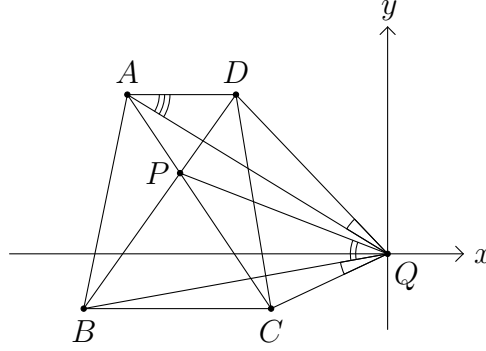
Problem 3.6. (IMO Shortlist 2007 G3/Baltic Way 1998 Problem 13) The diagonals of a trapezoid $ABCD$ intersect at point P . Point Q lies between the parallel lines BC and AD such that $\angle AQD = \angle CQB$, and line CD separates points P and Q . Prove that $\angle BQP = \angle DAQ$.



Analysis. This problem involves quite a lot of angles. Therefore, coordinate geometry should be the least suitable choice. But if one insists on using this, there is

nothing one can do except considering the slopes. As all angles involve point Q , we should set the origin at this point.

Solution.



Let $Q(0,0)$, $A(a,s)$, $B(b,t)$, $C(c,t)$, $D(d,s)$. Solving

$$AC: \quad y = \frac{s-t}{a-c}x + \frac{at-cs}{a-c}$$

and

$$BD: \quad y = \frac{s-t}{d-b}x + \frac{dt-bs}{d-b},$$

we have

$$\begin{aligned} x &= \frac{\frac{at-cs}{a-c} - \frac{dt-bs}{d-b}}{\frac{s-t}{d-b} - \frac{s-t}{a-c}} = \frac{(at-cs)(d-b) - (dt-bs)(a-c)}{(s-t)((a-c) - (d-b))} \\ &= \frac{(ab-cd)s + (cd-ab)t}{(s-t)(a+b-c-d)} = \frac{ab-cd}{a+b-c-d}, \\ y &= \frac{s-t}{a-c} \cdot \frac{ab-cd}{a+b-c-d} + \frac{at-cs}{a-c} = \frac{(ab-cd)(s-t) + (a+b-c-d)(at-cs)}{(a-c)(a+b-c-d)} \\ &= \frac{(ab-ac-bc+c^2)s + (cd+a^2-ac-ad)t}{(a-c)(a+b-c-d)} = \frac{(b-c)s + (a-d)t}{a+b-c-d}. \end{aligned}$$

That means $P\left(\frac{ab-cd}{a+b-c-d}, \frac{(b-c)s + (a-d)t}{a+b-c-d}\right)$.

We now find the slopes

$$m_{AQ} = \frac{s}{a}, \quad m_{DQ} = \frac{s}{d}, \quad m_{CQ} = \frac{t}{c}, \quad m_{BQ} = \frac{t}{b}, \quad m_{PQ} = \frac{(b-c)s + (a-d)t}{ab-cd}.$$

Now,

$$\begin{aligned}
& \angle AQD = \angle CQB \\
\Rightarrow & \tan \angle AQD = \tan \angle CQB \\
\Rightarrow & \frac{m_{AQ} - m_{DQ}}{1 + m_{AQ} \cdot m_{DQ}} = \frac{m_{CQ} - m_{BQ}}{1 + m_{CQ} \cdot m_{BQ}} \\
\Rightarrow & \frac{\frac{s}{a} - \frac{s}{d}}{1 + \frac{s}{a} \cdot \frac{s}{d}} = \frac{\frac{t}{c} - \frac{t}{b}}{1 + \frac{t}{c} \cdot \frac{t}{b}} \\
\Rightarrow & \frac{(d-a)s}{s^2 + ad} = \frac{(b-c)t}{t^2 + bc} \\
\Rightarrow & (d-a)st^2 + bc(d-a)s = (b-c)s^2t + ad(b-c)t \\
\Rightarrow & (b-c)s^2t + (a-d)st^2 + bc(a-d)s + ad(b-c)t = 0.
\end{aligned}$$

Next,

$$\begin{aligned}
& \angle BQP = \angle DAQ \\
\Leftrightarrow & \tan \angle BQP = \tan \angle DAQ \\
\Leftrightarrow & \frac{m_{BQ} - m_{PQ}}{1 + m_{BQ} \cdot m_{PQ}} = -m_{AQ} \\
\Leftrightarrow & \frac{\frac{t}{b} - \frac{(b-c)s + (a-d)t}{ab - cd}}{1 + \frac{t}{b} \cdot \frac{(b-c)s + (a-d)t}{ab - cd}} = -\frac{s}{a} \\
\Leftrightarrow & \frac{(ab - cd)t - b(b-c)s - b(a-d)t}{b(ab - cd) + (b-c)st + (a-d)t^2} = -\frac{s}{a} \\
\Leftrightarrow & -ab(b-c)s + ad(b-c)t = -b(ab - cd)s - (b-c)s^2t - (a-d)st^2 \\
\Leftrightarrow & (b-c)s^2t + (a-d)st^2 + bc(a-d)s + ad(b-c)t = 0.
\end{aligned}$$

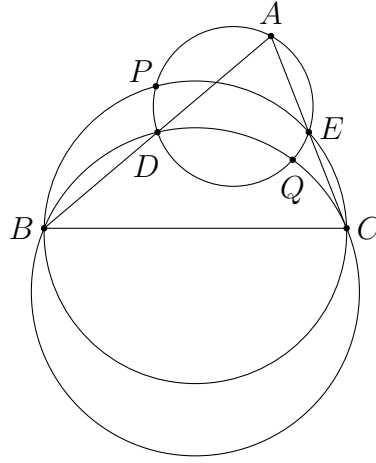
This is exactly the same as the above condition so that the angles are equal. $\square$

Remarks. As can be seen, using coordinates needs only a little skill. What we need is the patience to do the job. The only trick to handle the long expressions is to write in a certain order (like the degrees of s and t in the above solution).

4 Circles

Apart from angles, we always need to deal with circles in geometric problems. This is another tough work in coordinate geometry. In some of the previous problems, we demonstrate how to find the centre and equation of a circle. When we set up the coordinates, we tend to put the figure in a suitable place so that the centres of circles are easier to find. We shall also introduce other tricks to use the concyclic condition in the following problems.

Problem 4.1. (CGMO 2014 Problem 6) In acute $\triangle ABC$, $AB > AC$, D, E are the midpoints of AB, AC respectively. The circumcircle of $\triangle ADE$ and the circumcircle of $\triangle BCE$ meet at P (distinct from point E). The circumcircle of $\triangle ADE$ and the circumcircle of $\triangle BCD$ meet at Q (distinct from point D). Prove that $AP = AQ$.

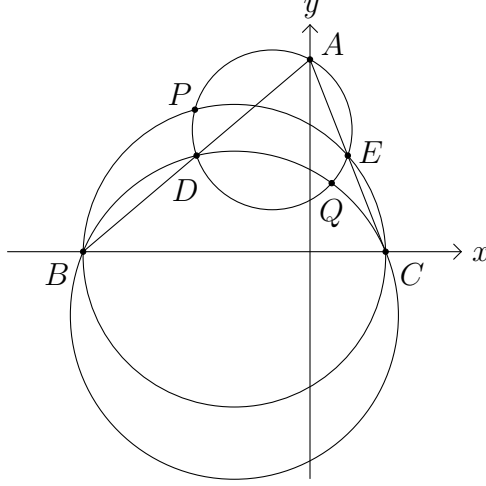


Analysis. There are 3 circles in this problem. However, it is still simple to use coordinate geometry. Indeed, using the standard setting, both BC and DE are parallel to the x -axis. Then the circumcentres of the triangles involved are easy to find. Also, we may use the symmetric technique. We demonstrate the routine steps to deal with this kind of problems.

Solution. Let $A(0, 2a)$, $B(2b, 0)$ and $C(2c, 0)$. Then $D(b, a)$ and $E(c, a)$. We first consider (ADE) . Its centre O has x -coordinate $\frac{b+c}{2}$. From

$$\begin{aligned}
 & OA = OD \\
 \Rightarrow & \left(\frac{b+c}{2}\right)^2 + (y-2a)^2 = \left(\frac{b+c}{2} - b\right)^2 + (y-a)^2 \\
 \Rightarrow & 4a^2 - 4ay = b^2 - b(b+c) + a^2 - 2ay \\
 \Rightarrow & y = \frac{3a^2 + bc}{2a},
 \end{aligned}$$

we get $O\left(\frac{b+c}{2}, \frac{3a^2+bc}{2a}\right)$.



Next, we find the centre O' of (BCE) . Its x -coordinate is $b+c$. From

$$\begin{aligned} O'E &= O'B \\ \Rightarrow (b+c-c)^2 + (y-a)^2 &= (b+c-2b)^2 + y^2 \\ \Rightarrow b^2 + a^2 - 2ay &= b^2 - 2bc + c^2 \\ \Rightarrow y &= \frac{a^2 + 2bc - c^2}{2a}, \end{aligned}$$

we get $O'\left(b+c, \frac{a^2 + 2bc - c^2}{2a}\right)$.

To find the second intersection point P of the two circles, we need not combine the equations of the two circles directly. Indeed, we shall use the fact that OO' is the perpendicular bisector of PE . It is easy to find

$$OO': \quad y = -\frac{2a^2 - bc + c^2}{a(b+c)}x + \frac{5a^2 + c^2}{2a}. \quad (1)$$

Then $m_{PE} = \frac{a(b+c)}{2a^2 - bc + c^2}$ and hence

$$PE: \quad y = \frac{a(b+c)}{2a^2 - bc + c^2}x + \frac{2a(a^2 - bc)}{2a^2 - bc + c^2}. \quad (2)$$

Solving (1) and (2), we have

$$\begin{aligned}
 x &= \frac{\frac{5a^2 + c^2}{2a} - \frac{2a(a^2 - bc)}{2a^2 - bc + c^2}}{\frac{2a^2 - bc + c^2}{a(b+c)} + \frac{a(b+c)}{2a^2 - bc + c^2}} = \frac{(b+c)(6a^4 - (b-7c)ca^2 - (b-c)c^3)}{2(4a^4 + (b^2 - 2bc + 5c^2)a^2 + (b-c)^2c^2)} \\
 &= \frac{(b+c)(a^2 + c^2)(6a^2 - bc + c^2)}{2(a^2 + c^2)(4a^2 + (b-c)^2)} = \frac{(b+c)(6a^2 - bc + c^2)}{2(4a^2 + (b-c)^2)}
 \end{aligned}$$

and

$$y = -\frac{2a^2 - bc + c^2}{a(b+c)} \cdot \frac{(b+c)(6a^2 - bc + c^2)}{2(4a^2 + (b-c)^2)} + \frac{5a^2 + c^2}{2a} = \frac{a(8a^2 + 5b^2 - 2bc + c^2)}{2(4a^2 + (b-c)^2)}.$$

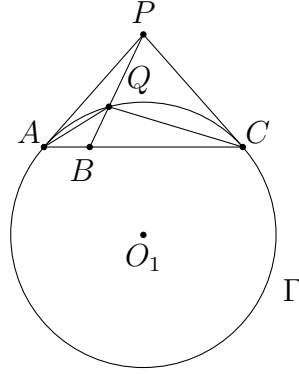
This is the midpoint of PE , so that $P\left(\frac{2(3b+c)a^2 - 2bc(b-c)}{4a^2 + (b-c)^2}, \frac{4a(a^2 + b^2)}{4a^2 + (b-c)^2}\right)$. Then we find

$$\begin{aligned}
 AP^2 &= \left(\frac{2(3b+c)a^2 - 2bc(b-c)}{4a^2 + (b-c)^2}\right)^2 + \left(\frac{4a(a^2 + b^2)}{4a^2 + (b-c)^2} - 2a\right)^2 \\
 &= \left(\frac{2(3b+c)a^2 - 2bc(b-c)}{4a^2 + (b-c)^2}\right)^2 + \left(\frac{2a(-2a^2 + b^2 + 2bc - c^2)}{4a^2 + (b-c)^2}\right)^2 \\
 &= \frac{16a^6 + (20b^2 - 8bc + 20c^2)a^4 + (4b^4 - 8b^3c + 24b^2c^2 - 8bc^3 + 4c^4)a^2 + 4b^2c^2(b-c)^2}{(4a^2 + (b-c)^2)^2}.
 \end{aligned}$$

No matter how clumsy this is, the only thing that we need is that the expression is symmetry in b and c . Therefore, we immediately obtain the result $AP = AQ$. $\square$

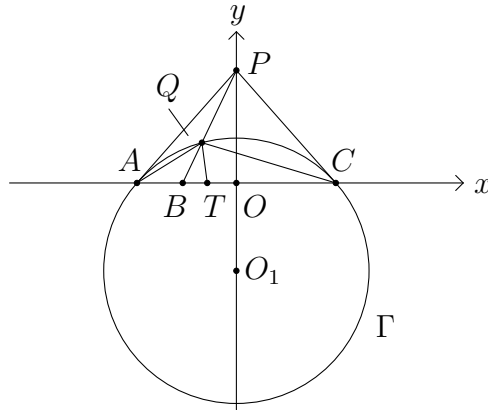
Remarks. There is no simple way to combine two quadratic equations to obtain the roots. If one of the intersection points of the two circles is known and we have the coordinates of the centres, then we usually use the method illustrated in this solution.

Problem 4.2. (IMO Shortlist 2003 G2) Three distinct points A, B , and C are fixed on a line in this order. Let Γ be a circle passing through A and C whose centre does not lie on the line AC . Denote by P the intersection of the tangents to Γ at A and C . Suppose Γ meets the segment PB at Q . Prove that the intersection of the bisector of $\angle AQC$ and the line AC does not depend on the choice of Γ .



Analysis. It seems that the problem involves angles and is not easy to be solved by coordinates. But the angle bisector theorem enables us to transform angles to ratio of sides. To prove the intersection point T is independent of Γ , we only need to prove $\frac{AT}{TC}$ is fixed (then the theorem can be used directly). We can take AC to be the x -axis so that the centre of Γ lies on the y -axis.

Solution.



Let $A(-1,0)$, $B(b,0)$, $C(1,0)$, and the centre of Γ be $O_1(0,a)$ with $a < 0$. By symmetry, P lies on the y -axis with y -coordinate $p > 0$. Recall the fact that P, A, O_1, C are concyclic (as $\angle PAO_1 + \angle PCO_1 = 180^\circ$). We apply the power of point theorem to get $AO \cdot CO = PO \cdot O_1O$. This means $1 \cdot 1 = p \cdot (-a)$ so that $p = -\frac{1}{a}$. The coordinates of P are $\left(0, -\frac{1}{a}\right)$. Clearly, the equation of PB is $y = \frac{1}{ab}x - \frac{1}{a}$. We shall write it in another form

$$PB: \quad x = aby + b \quad (1)$$

for convenience. Also,

$$\Gamma: \quad x^2 + y^2 - 2ay = 1. \quad (2)$$

To find $Q(q,r)$, we solve (1) and (2). These combine to give $(abr + b)^2 + r^2 - 2ar = 1$, i.e.

$$(a^2b^2 + 1)r^2 + 2a(b^2 - 1)r + (b^2 - 1) = 0.$$

There is no need to solve for r explicitly. This condition will be used afterwards.

Let the angle bisector of $\angle AQC$ meet the line AC at T . We are going to prove that $\frac{AT}{TC}$ is a constant, i.e. it can be expressed by b only. By the angle bisector theorem, we have

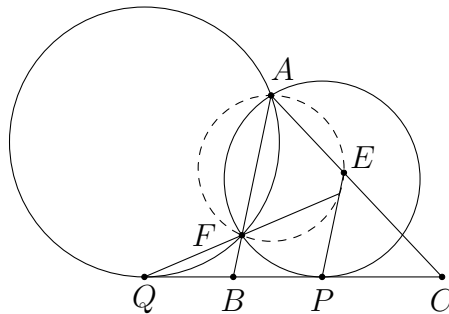
$$\begin{aligned}\frac{AT}{TC} &= \frac{AQ}{QC} = \sqrt{\frac{(q+1)^2 + r^2}{(q-1)^2 + r^2}} = \sqrt{\frac{(abr+b+1)^2 + r^2}{(abr+b-1)^2 + r^2}} \\ &= \sqrt{\frac{(a^2b^2+1)r^2 + 2ab(b+1)r + (b+1)^2}{(a^2b^2+1)r^2 + 2ab(b-1)r + (b-1)^2}} \\ &= \sqrt{\frac{(a^2b^2+1)r^2 + 2a(b^2-1)r + (b^2-1) + 2a(1+b)r + 2(1+b)}{(a^2b^2+1)r^2 + 2a(b^2-1)r + (b^2-1) + 2a(1-b)r + 2(1-b)}} \\ &= \sqrt{\frac{2a(1+b)r + 2(1+b)}{2a(1-b)r + 2(1-b)}} = \sqrt{\frac{1+b}{1-b}}.\end{aligned}$$

So we are done. $\square$

Remarks. In this solution, we use the angle bisector theorem and the power of point theorem. These theorems are simple but simplify our calculations a lot here.

Usually, we do not need to solve quadratic equations in coordinate geometry. The quadratic relation can be used as a whole. In this solution, we just need to convert higher powers of r to linear expressions. Also, this solution shows that the same argument can be applied to the second intersection of the line PB and Γ .

Problem 4.3. (IMO Shortlist 2008 G4) In an acute triangle ABC segments BE and CF are altitudes. Two circles passing through the points A and F are tangent to the line BC at the points P and Q so that B lies between C and Q . Prove that the lines PE and QF intersect on the circumcircle of triangle AEF .

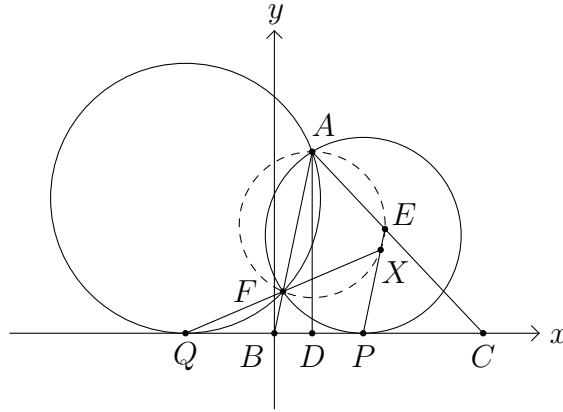


Analysis. To find the coordinates of P and Q , we do not need to find the equations of the circles explicitly. Indeed, a better method is to use powers. Note that there is

no need to use the standard setting for triangles since there is only a little symmetry in the figure. It is better to put B as the origin.

To prove that 4 points are concyclic, we choose not to find the equation of the circle passing through these points explicitly. The reason is that none of the perpendicular bisectors of the pairs of points are easy to find. Similar to the case of pure geometry, there are many ways to prove that 4 points are concyclic. For example, we may try to obtain a pair of equal angles by computing the slopes. One advantage is that we need not find the coordinates of $PE \cap QF$.

Solution.



Let $A(a, b)$, $B(0, 0)$, $C(1, 0)$ where $a, b > 0$. Then we have

$$AB: \quad y = \frac{b}{a}x,$$

$$AC: \quad y = \frac{b}{a-1}x - \frac{b}{a-1}$$

and

$$BE: \quad y = -\frac{a-1}{b}x,$$

$$CF: \quad y = -\frac{a}{b}x + \frac{a}{b}.$$

It is easy to get $E\left(\frac{b^2}{(a-1)^2 + b^2}, -\frac{b(a-1)}{(a-1)^2 + b^2}\right)$ and $F\left(\frac{a^2}{a^2 + b^2}, \frac{ab}{a^2 + b^2}\right)$. Let D be the foot of altitude from A . Its coordinates are $(a, 0)$. From

$$BP^2 = BQ^2 = BF \cdot BA = BD \cdot BC = a,$$

we get $P(\sqrt{a}, 0)$, $Q(-\sqrt{a}, 0)$. We compute the slopes

$$m_{PE} = -\frac{b(a-1)}{b^2 - \sqrt{a}((a-1)^2 + b^2)},$$

$$m_{QF} = \frac{ab}{a^2 + \sqrt{a}(a^2 + b^2)}.$$

Now, let $X = PE \cap QF$. Then

$$\begin{aligned}
& A, F, E, X \text{ are concyclic} \\
& \Leftrightarrow \angle(QF, PE) = \angle(AF, AE) \\
& \Leftrightarrow \frac{m_{QF} - m_{PE}}{1 + m_{QF} \cdot m_{PE}} = \frac{m_{AF} - m_{AE}}{1 + m_{AF} \cdot m_{AE}} \\
& \Leftrightarrow \frac{\frac{ab}{a^2 + \sqrt{a}(a^2 + b^2)} + \frac{b(a-1)}{b^2 - \sqrt{a}((a-1)^2 + b^2)}}{1 - \frac{ab}{a^2 + \sqrt{a}(a^2 + b^2)} \cdot \frac{b(a-1)}{b^2 - \sqrt{a}((a-1)^2 + b^2)}} = \frac{\frac{b}{a} - \frac{b}{a-1}}{1 + \frac{b}{a} \cdot \frac{b}{a-1}} \\
& \Leftrightarrow \frac{ab(b^2 - \sqrt{a}((a-1)^2 + b^2)) + b(a-1)(a^2 + \sqrt{a}(a^2 + b^2))}{(a^2 + \sqrt{a}(a^2 + b^2))(b^2 - \sqrt{a}((a-1)^2 + b^2)) - ab^2(a-1)} = \frac{b(a-1) - ab}{a(a-1) + b^2} \\
& \Leftrightarrow \frac{(ab^2 + a^3 - a^2) + \sqrt{a}(a^2 - b^2 - a)}{(-a^5 + 2a^4 - a^3 - 2a^3b^2 + 2a^2b^2 - ab^4) + \sqrt{a}(b^4 - a^2(a-1)^2)} = -\frac{1}{a^2 - a + b^2} \\
& \Leftrightarrow -\frac{(ab^2 + a^3 - a^2) + \sqrt{a}(a^2 - b^2 - a)}{((ab^2 + a^3 - a^2) + \sqrt{a}(a^2 - b^2 - a))(a^2 - a + b^2)} = -\frac{1}{a^2 - a + b^2}.
\end{aligned}$$

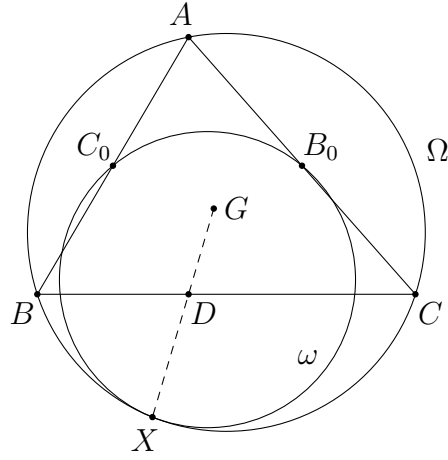
This is obvious. □

Remarks. Why do we know the factorization of the denominator of the last expression? this is simply because if we are correct, then the numerator must be cancelled out and the other factor must be the same as the right side. (You can write it down without verification if you are sure that you make no mistake!)

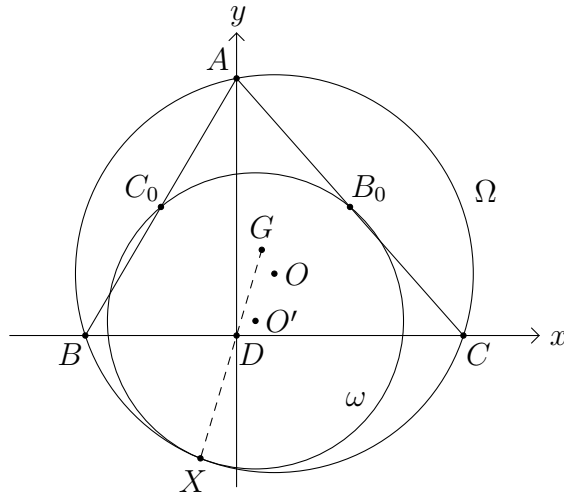
Problem 4.4. (IMO Shortlist 2011 G4) Let ABC be an acute triangle with circumcircle Ω . Let B_0 be the midpoint of AC and let C_0 be the midpoint of AB . Let D be the foot of the altitude from A , and let G be the centroid of the triangle ABC . Let ω be a circle through B_0 and C_0 that is tangent to the circle Ω at a point $X \neq A$. Prove that the points D, G , and X are collinear.

Analysis. Those who would choose to use coordinate geometry are probably weak in geometry. The only geometric fact that those would know and is useful for this problem is that the centres of the two circles and point X are collinear. However, this does not provide an obvious way to find the coordinates of X .

Note that there are only two circles passing through B_0, C_0 and tangent to Ω . The tangential point of one of them is A (which is easy to prove or can be guessed from the question), which guarantees that point X is uniquely determined. From this, we may try to do backward, i.e. find X by using one of the intersection points of DG and Ω , and claim that it is the same as the given X .



Solution.



Let $A(0, 2a)$, $B(2b, 0)$ and $C(2c, 0)$ where $a, c > 0 > b$ since $\triangle ABC$ is acute. WLOG we may further assume $AB \leq AC$ so that $b + c \geq 0$. Then $D(0, 0)$, $C_0(b, a)$, $B_0(c, a)$ and $G\left(\frac{2(b+c)}{3}, \frac{2a}{3}\right)$. Let O be the circumcentre of $\triangle ABC$. Its x -coordinate is $b + c$. Also,

$$\begin{aligned}
 & OA = OB \\
 \Rightarrow & (b+c)^2 + (y-2a)^2 = (b-c)^2 + y^2 \\
 \Rightarrow & 2bc + 4a^2 - 4ay = -2bc \\
 \Rightarrow & y = \frac{a^2 + bc}{a}.
 \end{aligned}$$

This gives $O\left(b+c, \frac{a^2+bc}{a}\right)$. From this, we get

$$\Omega: \quad x^2 + y^2 - 2(b+c)x - \frac{2(a^2+bc)}{a}y + 4bc = 0. \quad (1)$$

Next,

$$DG: \quad y = \frac{a}{b+c}x. \quad (2)$$

Putting (2) into (1), we solve $x^2 + \frac{a^2}{(b+c)^2}x^2 - 2(b+c)x - \frac{2(a^2+bc)}{b+c}x + 4bc = 0$. This is the same as

$$(a^2 + b^2 + 2bc + c^2)x^2 - 2(b+c)(a^2 + b^2 + 3bc + c^2)x + 4bc(b+c)^2 = 0.$$

Noting

$$\begin{aligned} \Delta &= 4(b+c)^2(a^2 + b^2 + 3bc + c^2)^2 - 16bc(b+c)^2(a^2 + b^2 + 2bc + c^2) \\ &= 4(b+c)^2(a^4 + 2(b^2 + bc + c^2)a^2 + (b^4 + 2b^3c + 3b^2c^2 + 2bc^3 + c^4)) \\ &= 4(b+c)^2(a^2 + b^2 + bc + c^2)^2, \end{aligned}$$

we easily obtain the two solutions $x = 2(b+c)$ or $\frac{2bc(b+c)}{a^2 + b^2 + 2bc + c^2}$. Since X should lie on the ray GD and hence has negative coordinates, we should reject the solution $2(b+c) > 0$ and focus on $x = \frac{2bc(b+c)}{a^2 + b^2 + 2bc + c^2}$. Therefore, we define

$$X' \left(\frac{2bc(b+c)}{a^2 + b^2 + 2bc + c^2}, \frac{2abc}{a^2 + b^2 + 2bc + c^2} \right).$$

From our discussion, as X is uniquely determined, it suffices to show that the circumcircle of $\triangle B_0C_0X'$ is tangent to Ω , since this implies $X' = X$ and hence X, D, G are collinear.

Let O' be the circumcentre of $\triangle B_0C_0X'$. The x -coordinate of O' is $\frac{b+c}{2}$. Next, the condition $O'C_0 = O'X'$ gives

$$\left(\frac{b-c}{2}\right)^2 + (y-a)^2 = \left(\frac{b+c}{2} - \frac{2bc(b+c)}{a^2 + b^2 + 2bc + c^2}\right)^2 + \left(y - \frac{2abc}{a^2 + b^2 + 2bc + c^2}\right)^2.$$

Directly expanding the things may not be so easy. Indeed, we rewrite this as

$$(y-a)^2 - \left(y - \frac{2abc}{a^2 + b^2 + 2bc + c^2}\right)^2 = \left(\frac{b+c}{2} - \frac{2bc(b+c)}{a^2 + b^2 + 2bc + c^2}\right)^2 - \left(\frac{b-c}{2}\right)^2.$$

Both sides can be factorized. For the left-hand side, we have

$$\begin{aligned} &(y-a)^2 - \left(y - \frac{2abc}{a^2 + b^2 + 2bc + c^2}\right)^2 \\ &= \left(\frac{2abc}{a^2 + b^2 + 2bc + c^2} - a\right) \left(2y - a - \frac{2abc}{a^2 + b^2 + 2bc + c^2}\right) \\ &= \frac{-a(a^2 + b^2 + c^2)}{a^2 + b^2 + 2bc + c^2} \cdot \left(2y - \frac{a(a^2 + b^2 + 4bc + c^2)}{a^2 + b^2 + 2bc + c^2}\right). \end{aligned}$$

For the right-hand side, we have

$$\begin{aligned}
& \left(\frac{b+c}{2} - \frac{2bc(b+c)}{a^2+b^2+2bc+c^2} \right)^2 - \left(\frac{b-c}{2} \right)^2 \\
&= \left(c - \frac{2bc(b+c)}{a^2+b^2+2bc+c^2} \right) \left(b - \frac{2bc(b+c)}{a^2+b^2+2bc+c^2} \right) \\
&= \frac{c(a^2-b^2+c^2)}{a^2+b^2+2bc+c^2} \cdot \frac{b(a^2+b^2-c^2)}{a^2+b^2+2bc+c^2}.
\end{aligned}$$

Equating these, we obtain

$$\begin{aligned}
y &= \frac{bc(a^2-b^2+c^2)(a^2+b^2-c^2)}{-2a(a^2+b^2+c^2)(a^2+b^2+2bc+c^2)} + \frac{a(a^2+b^2+4bc+c^2)}{2(a^2+b^2+2bc+c^2)} \\
&= \frac{a^6 + (2b^2+3bc+2c^2)a^4 + (b^4+4b^3c+2b^2c^2+4bc^3+c^4)a^2 + bc(b^2-c^2)^2}{2a(a^2+b^2+c^2)(a^2+b^2+2bc+c^2)} \\
&= \frac{a^4 + (b^2+bc+c^2)a^2 + bc(b-c)^2}{2a(a^2+b^2+c^2)}.
\end{aligned}$$

(Again, it is a good practice to test whether the factors can be eliminated.) This gives $O' \left(\frac{b+c}{2}, \frac{a^4 + (b^2+bc+c^2)a^2 + bc(b-c)^2}{2a(a^2+b^2+c^2)} \right)$. It remains to check O, O', X' are collinear. Firstly,

$$OX': \quad y = \frac{a^4 + (b^2+bc+c^2)a^2 + bc(b+c)^2}{a(b+c)(a^2+b^2+c^2)}x - \frac{2b^2c^2}{a(a^2+b^2+c^2)}.$$

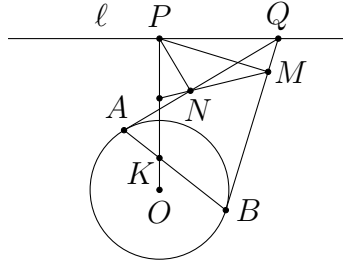
We now show that the coordinates of O' satisfy this equation:

$$\begin{aligned}
& \frac{a^4 + (b^2+bc+c^2)a^2 + bc(b+c)^2}{a(b+c)(a^2+b^2+c^2)} \cdot \frac{b+c}{2} - \frac{2b^2c^2}{a(a^2+b^2+c^2)} \\
&= \frac{a^4 + (b^2+bc+c^2)a^2 + bc(b-c)^2}{2a(a^2+b^2+c^2)}.
\end{aligned}$$

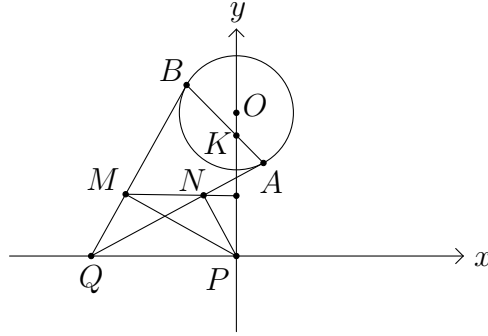
This completes the proof. □

Problem 4.5. (CSMO 2005 Problem 2) A circle O (with centre O) and a line ℓ do not intersect. Construct $OP \perp \ell$ where P is the foot of perpendicular. Let Q be an arbitrary point on ℓ (distinct from P). Construct two tangents QA and QB from Q to circle O , where A and B are the contact points. AB and OP intersect at K . Construct $PM \perp QB$ and $PN \perp QA$ where M and N are the feet of perpendiculars. Prove that the line MN bisects the segment KP .

Analysis. The first priority is to put the centre of the circle on the y -axis. Also, noting that most of the lines pass through point P , we set the origin at this point.



Solution.



Let $P(0,0)$, $O(0,1)$, $Q(a,0)$ and the equation of the circle be $x^2 + (y - 1)^2 = r^2$. Consider the family of lines through Q . Each line has the form $y = mx - ma$. To find the tangents from Q to $\odot O$, the equation obtained from combining the two equations has a multiple root.

$$\Leftrightarrow \begin{aligned} & x^2 + (mx - ma - 1)^2 = r^2 \\ & (m^2 + 1)x^2 - 2m(ma + 1)x + ((ma + 1)^2 - r^2) = 0. \end{aligned} \quad (1)$$

That means we need $\Delta = 4m^2(ma + 1)^2 - 4(m^2 + 1)((ma + 1)^2 - r^2) = 0$. This means $-(ma + 1)^2 + (m^2 + 1)r^2 = 0$, i.e.

$$(r^2 - a^2)m^2 - 2am + (r^2 - 1) = 0. \quad (2)$$

The two roots correspond to the slopes of QA and QB . Hence, we have

$$m_{QA} + m_{QB} = \frac{2a}{r^2 - a^2}, \quad m_{QA} \cdot m_{QB} = \frac{r^2 - 1}{r^2 - a^2}.$$

To find K , the quickest method is to note that K is the inversion point of P w.r.t. $\odot O$ (indeed ℓ is the polar line of K w.r.t. $\odot O$), hence $OP \cdot OK = r^2$. Alternatively, we can still find it using its definition. The coordinates of A, B can be found from (1). Note that the quadratic equation has a multiple root so we need not consider the constant term. We easily get

$$x = \frac{m(ma + 1)}{m^2 + 1}, \quad y = m(x - a) = \frac{m(m - a)}{m^2 + 1}.$$

As $AB \perp QO$, we have $m_{AB} = a$. Thus,

$$AB: \quad y = ax - \frac{ma(ma+1)}{m^2+1} + \frac{m(m-a)}{m^2+1}.$$

Here m can be either m_{QA} or m_{QB} . We now use (2) to simplify this expression. Indeed,

$$\begin{aligned} y &= ax + \frac{(1-a^2)m^2 - 2am}{m^2+1} = ax + \frac{(1-a^2)m^2 - (r^2-a^2)m^2 - (r^2-1)}{m^2+1} \\ &= ax + \frac{(1-r^2)(m^2+1)}{m^2+1} = ax + 1 - r^2. \end{aligned}$$

As K lies on the y -axis, we find $K(0, 1-r^2)$.

Next, we solve

$$PM: \quad y = -\frac{1}{m_{QB}}x,$$

$$QB: \quad y = m_{QB}x - m_{QB}a$$

to get $M\left(\frac{m_{QB}^2a}{m_{QB}^2+1}, -\frac{m_{QB}a}{m_{QB}^2+1}\right)$ and hence $N\left(\frac{m_{QA}^2a}{m_{QA}^2+1}, -\frac{m_{QA}a}{m_{QA}^2+1}\right)$. This gives

$$\begin{aligned} m_{MN} &= \frac{-m_{QB}(m_{QA}^2+1) + m_{QA}(m_{QB}^2+1)}{m_{QB}^2(m_{QA}^2+1) - m_{QA}^2(m_{QB}^2+1)} = \frac{(m_{QB} - m_{QA})(m_{QA} \cdot m_{QB} - 1)}{m_{QB}^2 - m_{QA}^2} \\ &= \frac{m_{QA} \cdot m_{QB} - 1}{m_{QA} + m_{QB}} = \frac{\frac{r^2-1}{r^2-a^2} - 1}{\frac{2a}{r^2-a^2}} = \frac{a^2-1}{2a}. \end{aligned}$$

We need to prove that the equation of MN is

$$y = \frac{a^2-1}{2a}x + \frac{1-r^2}{2},$$

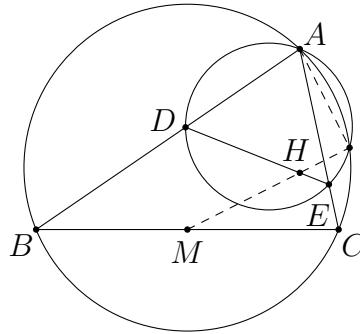
as this implies the y -intercept is $\frac{1}{2}(1-r^2)$, which is the midpoint of KP . Using either $m = m_{QA}$ or m_{QB} , we have

$$\begin{aligned} -\frac{ma}{m^2+1} &= \frac{a^2-1}{2a} \cdot \frac{m^2a}{m^2+1} + \frac{1-r^2}{2} \\ \Leftrightarrow -2ma &= (a^2-1)m^2 + (1-r^2)(m^2+1) \\ \Leftrightarrow 0 &= (a^2-r^2)m^2 + 2am + (1-r^2). \end{aligned}$$

This is exactly the same as (2). Hence, we are done. $\square$

Remarks. Similar to [problem 4.2](#), we do not need to find all coordinates and equations explicitly. Here, we only express M, N as the slopes of QA, QB , as we know a lot of facts about these slopes from the quadratic equation. Also, we do not follow the steps to find MN and then deduce its y -intercept. It is much simpler if we see the other way round.

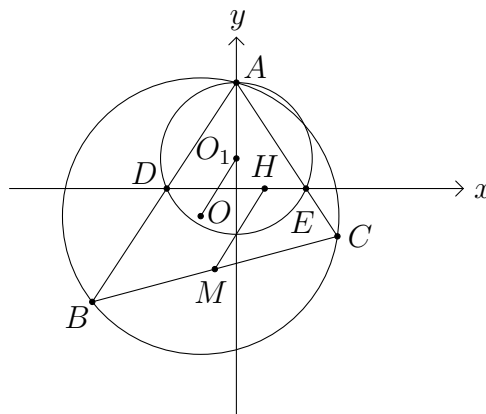
Problem 4.6. (IMO Shortlist 2005 G5) Let $\triangle ABC$ be an acute-angled triangle with $AB \neq AC$. Let H be the orthocentre of $\triangle ABC$, and let M be the midpoint of the side BC . Let D be a point on the side AB and E a point on the side AC such that $AE = AD$ and the points D, H, E are on the same line. Prove that the line HM is perpendicular to the common chord of the circumscribed circles of $\triangle ABC$ and $\triangle ADE$.



Analysis. The equation of the common chord of two circles (the radical axis) can be found by taking the difference of the equations of these circles. Instead of proving the two lines are perpendicular, we can simply prove that HM is parallel to the line joining the centres of the circles, as the radical axis is perpendicular to this line.

One of the difficulties in this problem is to determine the coordinates of D and E . We try to use some geometric way to interpret the condition $AD = AE$. This holds iff DE is perpendicular to the internal angle bisector of $\angle A$. Therefore, we try to take this angle bisector to be the y -axis to simplify a lot of calculations.

Solution.



Let $A(0, a)$, $D(-1, 0)$, $E(1, 0)$, $H(b, 0)$. From

$$\begin{aligned} AB: & \quad y = ax + a, \\ AC: & \quad y = -ax + a \end{aligned}$$

and

$$\begin{aligned} CH: \quad y &= -\frac{1}{a}x + \frac{b}{a}, \\ BH: \quad y &= \frac{1}{a}x - \frac{b}{a}, \end{aligned}$$

we obtain $B\left(-\frac{a^2+b}{a^2-1}, -\frac{a(1+b)}{a^2-1}\right)$ and $C\left(\frac{a^2-b}{a^2-1}, \frac{a(b-1)}{a^2-1}\right)$ so that we have the coordinates $M\left(-\frac{b}{a^2-1}, -\frac{a}{a^2-1}\right)$.

Let O and O_1 be the circumcentres of $\triangle ABC$ and $\triangle ADE$ respectively. Firstly, let further G be the centroid of $\triangle ABC$, whose coordinates are $\left(-\frac{2b}{3(a^2-1)}, \frac{a(a^2-3)}{3(a^2-1)}\right)$.

Using the fact $HO : OG = 3 : (-1)$, we find

$$O = \left(\frac{3\left(-\frac{2b}{3(a^2-1)}\right) + (-1)(b)}{3 + (-1)}, \frac{3\left(\frac{a(a^2-3)}{3(a^2-1)}\right) + (-1)(0)}{3 + (-1)} \right) = \left(-\frac{b(a^2+1)}{2(a^2-1)}, \frac{a(a^2-3)}{2(a^2-1)} \right).$$

(One may also check the fact that $m_{AO} = -m_{AH}$ as the two lines are symmetric in the A -angle bisector.)

For O_1 , it must lie on the y -axis. Also,

$$\begin{aligned} AO_1 &= DO_1 \\ \Rightarrow (a-y)^2 &= y^2 + 1 \\ \Rightarrow y &= \frac{a^2-1}{2a}. \end{aligned}$$

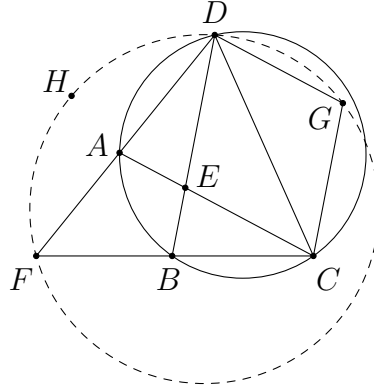
This gives $O_1\left(0, \frac{a^2-1}{2a}\right)$. Now,

$$\begin{aligned} m_{MH} &= \frac{\frac{a}{a^2-1}}{b + \frac{b}{a^2-1}} = \frac{1}{ab}, \\ m_{OO_1} &= \frac{\frac{a^2-1}{2a} - \frac{a(a^2-3)}{2(a^2-1)}}{\frac{b(a^2+1)}{2(a^2-1)}} = \frac{(a^2-1)^2 - a^2(a^2-3)}{ab(a^2+1)} = \frac{1}{ab}. \end{aligned}$$

This shows MH and OO_1 are parallel. As OO_1 is perpendicular to the common chord of (ABC) and (ADE) , the assertion follows readily. $\square$

Remarks. This construction is a bit odd. But it turns out that it is very useful to deal with this problem. The only trick is that we always need to be flexible!

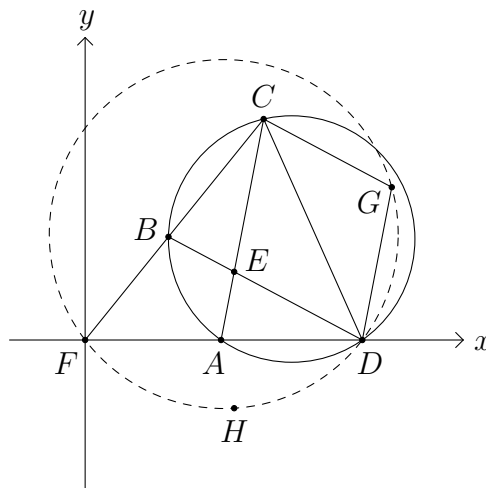
Problem 4.7. (IMO Shortlist 2012 G2) Let $ABCD$ be a cyclic quadrilateral whose diagonals AC and BD meet at E . The extensions of the sides AD and BC beyond A and B meet at F . Let G be the point such that $ECGD$ is a parallelogram, and let H be the image of E under reflection in AD . Prove that D, H, F, G are concyclic.



Analysis. Apart from triangles, we usually need to handle cyclic quadrilaterals. In that case, extending the standard setting of a triangle does not seem to simplify the calculation. In particular, the coordinates of the fourth point usually involve radicals and the equation of the circle is not simple. If we put $(ABCD)$ as the unit circle centered at the origin, then the equation of the circle is much simplified, but this may be even worse as we cannot help dealing with square roots.

Here, we shall introduce another way to assign a condition to concyclic points, i.e. putting the intersection of the diagonals or opposite sides as the origin. Surprisingly, the power condition is rather simple and there is no need to deal with square roots. In addition, we set the x -axis at AD so that the reflection can be easily done.

Solution.



Let $F(0,0)$, $A(a,0)$, $D(d,0)$, $B(b,mb)$, $C(c,mc)$ so that the equation of BC is $y = mx$. The concyclic condition is equivalent to $FB \cdot FC = FA \cdot FD$. This means

$$\sqrt{b^2 + m^2 b^2} \cdot \sqrt{c^2 + m^2 c^2} = ad, \text{ i.e.}$$

$$bc(m^2 + 1) = ad. \quad (1)$$

Here, we have used the fact that bc and ad are positive from the configuration. To find E , we solve

$$\begin{aligned} AC : \quad y &= \frac{mc}{c-a}x - \frac{mac}{c-a}, \\ BD : \quad y &= \frac{mb}{b-d}x - \frac{mbd}{b-d} \end{aligned}$$

to get $E\left(\frac{ac(b-d) - bd(c-a)}{ab-cd}, \frac{mbc(a-d)}{ab-cd}\right)$. Reflecting in the x -axis, we immediately obtain $H\left(\frac{ac(b-d) - bd(c-a)}{ab-cd}, -\frac{mbc(a-d)}{ab-cd}\right)$.

It is also easy to find the coordinates of G . Since the diagonals of a parallelogram bisect each other, we get $G = C + D - E = \left(\frac{cd(a+b-c-d)}{ab-cd}, \frac{mcd(b-c)}{ab-cd}\right)$.

For any point $X(\alpha, \beta)$, let $O_X\left(\frac{d}{2}, y\right)$ be the circumcentre of $\triangle FDX$. From

$$\begin{aligned} O_X F &= O_X X \\ \Rightarrow \quad \frac{d^2}{4} + y^2 &= \left(\frac{d}{2} - \alpha\right)^2 + (y - \beta)^2 \\ \Rightarrow \quad 0 &= \alpha^2 - d\alpha + \beta^2 - 2\beta y \\ \Rightarrow \quad y &= \frac{\alpha^2 - d\alpha + \beta^2}{2\beta}, \end{aligned}$$

we get $O_X\left(\frac{d}{2}, \frac{\alpha^2 - d\alpha + \beta^2}{2\beta}\right)$. Note that D, H, F, G are concyclic iff the circumcentre of $\triangle FDG$ coincides with the circumcentre of $\triangle FDH$. For convenience, we write $G(g_x, g_y)$ and $H(h_x, h_y)$. From above, we have to prove

$$\frac{g_x^2 - dg_x + g_y^2}{2g_y} = \frac{h_x^2 - dh_x + h_y^2}{2h_y}. \quad (2)$$

Note that

$$\frac{g_x^2 - dg_x}{g_y} = \frac{g_x(g_x - d)}{g_y} = \frac{g_x d(c-a)(b-c)}{g_y(ab-cd)} = \frac{d(c-a)(a+b-c-d)}{m(ab-cd)}$$

and

$$\frac{h_x^2 - dh_x}{h_y} = \frac{h_x(h_x - d)}{h_y} = \frac{h_x c(a-d)(b-d)}{h_y(ab-cd)} = -\frac{(b-d)(ac(b-d) - bd(c-a))}{mb(ab-cd)}.$$

Hence, we have

$$\begin{aligned}
\frac{g_x^2 - dg_x}{g_y} - \frac{h_x^2 - dh_x}{h_y} &= \frac{d(c-a)(a+b-c-d)}{m(ab-cd)} + \frac{(b-d)(ac(b-d) - bd(c-a))}{mb(ab-cd)} \\
&= \frac{bd(c-a)((a-c) + (b-d)) + (b-d)(ac(b-d) - bd(c-a))}{mb(ab-cd)} \\
&= \frac{-bd(a-c)^2 + ac(b-d)^2}{mb(ab-cd)}.
\end{aligned}$$

Recall we need to show (2). Indeed,

$$\begin{aligned}
&\frac{g_x^2 - dg_x + g_y^2}{2g_y} = \frac{h_x^2 - dh_x + h_y^2}{2h_y} \\
\Leftrightarrow &\frac{g_x^2 - dg_x}{g_y} - \frac{h_x^2 - dh_x}{h_y} = h_y - g_y \\
\Leftrightarrow &\frac{-bd(a-c)^2 + ac(b-d)^2}{mb(\textcolor{red}{ab-cd})} = -\frac{mbc(a-d)}{\textcolor{red}{ab-cd}} - \frac{mcd(b-c)}{\textcolor{red}{ab-cd}} \\
\Leftrightarrow &-bd(a-c)^2 + ac(b-d)^2 = -m^2b^2c(a-d) - m^2bcd(b-c) \\
\Leftrightarrow &-bd(a^2 + c^2) + ac(b^2 + d^2) = -m^2bc(ab-cd) \\
\Leftrightarrow &(ab-cd)(bc-ad) = -m^2bc(ab-cd) \\
\Leftrightarrow &(ab-cd)(bc(m^2+1) - ad) = 0.
\end{aligned}$$

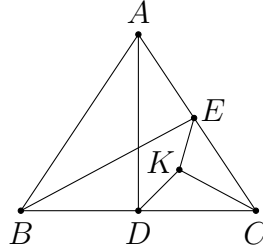
This is true from (1). □

Remarks. For problems involving a cyclic quadrilateral $ABCD$, setting $AB \cap CD$, $AC \cap BD$ or $AD \cap BC$ as the origin is an extremely useful technique. The power relation gives a simple condition on the variables, and this usually retains the symmetry of the concyclic points.

5 Incentre and Excentre

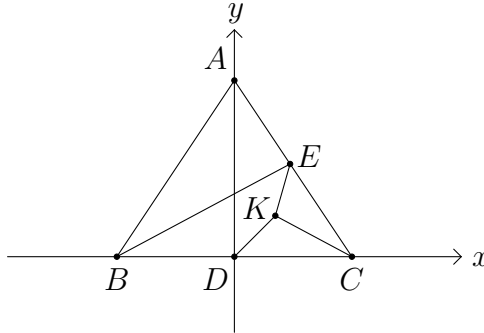
In this section, we shall deal with a number of problems related to the incentre and incircle, and also the excircles. It is thought that coordinate geometry is difficult to be applied to incentre as angle bisectors of given angles have unfriendly expressions. However, if we set the coordinates appropriately, such problems can still be solved in an easy manner (relatively).

Problem 5.1. (IMO Shortlist 2009 G1/Problem 4) Let ABC be a triangle with $AB = AC$. The angle bisectors of $\angle CAB$ and $\angle ABC$ meet the sides BC and CA at D and E , respectively. Let K be the incentre of triangle ADC . Suppose that $\angle BEK = 45^\circ$. Find all possible values of $\angle CAB$.



Analysis. We first begin this section with a simple case, where the given triangle is an isosceles one. In that case, we usually set the origin at the midpoint of the base. Then all points can be expressed in one variable only. This is essential as we are going to solve an equation in the last step.

Solution.



Let $D(0,0)$, $B(-1,0)$, $C(1,0)$, $A(0,a)$ where $a > 0$. Using the angle bisector theorem, we have

$$\frac{AE}{EC} = \frac{AB}{BC} = \frac{\sqrt{a^2 + 1}}{2}.$$

Then we can apply the sectional formula to obtain $E\left(\frac{\sqrt{a^2 + 1}}{2 + \sqrt{a^2 + 1}}, \frac{2a}{2 + \sqrt{a^2 + 1}}\right)$.

To find K , we choose two of the internal angle bisectors and find their intersection. The simplest one is DK . As $\angle CDK = \angle KDA = 45^\circ$, the slope of DK is $\tan 45^\circ = 1$ and hence its equation is $y = x$. Next, CK should be easier to find than AK . One reason is that CK and BE are symmetric in the y -axis (as $\triangle ABC$ is isosceles). Then we can use the equation

$$BE: \quad y = \frac{a}{1 + \sqrt{a^2 + 1}}x + \frac{a}{1 + \sqrt{a^2 + 1}}$$

to find

$$CK: \quad y = -\frac{a}{1 + \sqrt{a^2 + 1}}x + \frac{a}{1 + \sqrt{a^2 + 1}}.$$

From CK and DK , we easily get $K\left(\frac{a}{a+1+\sqrt{a^2+1}}, \frac{a}{a+1+\sqrt{a^2+1}}\right)$. Then

$$m_{EK} = \frac{2a(a+1+\sqrt{a^2+1}) - a(2+\sqrt{a^2+1})}{\sqrt{a^2+1} \cdot (a+1+\sqrt{a^2+1}) - a(2+\sqrt{a^2+1})} = \frac{2a^2 + a\sqrt{a^2+1}}{a^2 - 2a + 1 + \sqrt{a^2+1}},$$

and so

$$\begin{aligned} \angle BEK &= 45^\circ \\ \Leftrightarrow \quad \tan \angle BEK &= 1 \\ \Leftrightarrow \quad \frac{m_{KE} - m_{BE}}{1 + m_{KE} \cdot m_{BE}} &= 1 \\ \Leftrightarrow \quad \frac{2a^2 + a\sqrt{a^2+1}}{a^2 - 2a + 1 + \sqrt{a^2+1}} - \frac{a}{1 + \sqrt{a^2+1}} \\ &= 1 + \frac{2a^2 + a\sqrt{a^2+1}}{a^2 - 2a + 1 + \sqrt{a^2+1}} \cdot \frac{a}{1 + \sqrt{a^2+1}} \\ \Leftrightarrow \quad (2a^2 + a\sqrt{a^2+1})(1 + \sqrt{a^2+1}) - a(a^2 - 2a + 1 + \sqrt{a^2+1}) \\ &= (a^2 - 2a + 1 + \sqrt{a^2+1})(1 + \sqrt{a^2+1}) + a(2a^2 + a\sqrt{a^2+1}) \\ \Leftrightarrow \quad 4a^2 + 2a^2\sqrt{a^2+1} &= (2a^3 + 2a^2 - 2a + 2) + (2a^2 - 2a + 2)\sqrt{a^2+1} \\ \Leftrightarrow \quad a^3 - a^2 - a + 1 &= (a-1)\sqrt{a^2+1} \\ \Leftrightarrow \quad (a-1)(a^2 - 1) &= (a-1)\sqrt{a^2+1}. \end{aligned}$$

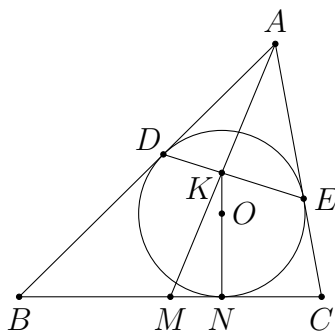
Clearly, $a = 1$ is a solution, and there is no solution for $a < 1$. For $a > 1$, this holds iff $(a^2 - 1)^2 = a^2 + 1$, which is the same as $a^4 - 3a^2 = 0$. The solution with $a > 1$ is $a = \sqrt{3}$. The two values of a correspond to $\angle BAC = 90^\circ$ and 60° respectively. As the steps are reversible, no checking is needed. $\square$

Remarks. Though the solution involves square root signs, one can see that it does not really cause troubles in the calculation. Indeed, we assume the whole sign as a variable and group the terms systematically. Since only one variable is involved and the degree is not so high, such factorizations are not difficult to see. We do not even

have to guess the answer before solving the equation (though the angle must be some special angles).

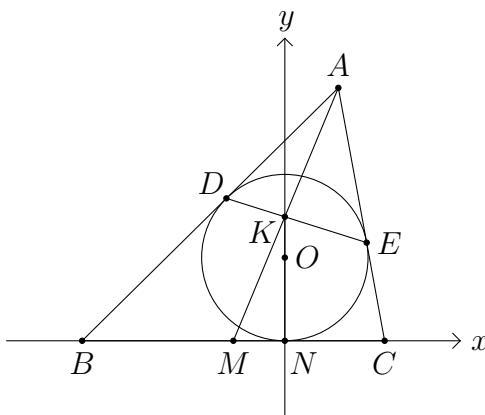
In this case, the angle bisectors are not hard to find. The only possible trick we have used is the angle bisector theorem. However, this may not help a lot in the more general case.

Problem 5.2. (NMO 2010 Problem 6) $\odot O$ is the incircle of $\triangle ABC$ where D, E, N are the contact points. The extension of NO meets DE at K . The extension of AK meets BC at M . Prove that M is the midpoint of BC .



Analysis. In this example, we introduce a useful setting for triangle geometry involving the incentre. If we first define the coordinates of A, B, C and find the coordinates of the incentre, then this is a tedious task and one would probably end up with some square roots in the expression, like what we have done in [problem 5.1](#). In view of this, our idea is to define the coordinates of the incentre and two of the vertices first, and then find the coordinates of the remaining vertex (sometimes only the equations of the sidelines are needed).

Solution.



Let $N(0,0)$, $O(0,1)$, $B(b,0)$, $C(c,0)$. From the angle bisector, (where we use x to denote the x -axis)

$$m_{AB} = \tan \angle(x, AB) = \tan(2\angle(x, BO)) = \frac{2 \tan \angle(x, BO)}{1 - \tan^2 \angle(x, BO)} = \frac{2m_{BO}}{1 - m_{BO}^2}.$$

This gives $m_{AB} = -\frac{2b}{b^2 - 1}$ so that

$$AB: \quad y = -\frac{2b}{b^2 - 1}x + \frac{2b^2}{b^2 - 1}. \quad (1)$$

As directed angles are used, the same holds for AC and hence

$$AC: \quad y = -\frac{2c}{c^2 - 1}x + \frac{2c^2}{c^2 - 1}. \quad (2)$$

Solving (1) and (2), we get $A\left(\frac{b+c}{bc+1}, \frac{2bc}{bc+1}\right)$. To find the coordinates of D , instead of solving the equations of AB and $\odot O$, we use the geometric fact that $BO \perp DN$. This easily implies $m_{DN} = b$ and hence

$$DN: \quad y = bx. \quad (3)$$

Solving (1) and (3), we obtain $D\left(\frac{2b}{b^2+1}, \frac{2b^2}{b^2+1}\right)$. Thus, we have $E\left(\frac{2c}{c^2+1}, \frac{2c^2}{c^2+1}\right)$ by symmetry. Then

$$DE: \quad y = -\frac{b+c}{bc-1}x + \frac{2bc}{bc-1}.$$

(An indirect way to find the slope of DE is to use the fact that $AO \perp DE$.) As K lies on the y -axis, we have $K\left(0, \frac{2bc}{bc-1}\right)$. Let $M'\left(\frac{b+c}{2}, 0\right)$ be the midpoint of BC . It suffices to prove A, K, M' are collinear. We find that

$$KM': \quad y = -\frac{4bc}{(bc-1)(b+c)}x + \frac{2bc}{bc-1}.$$

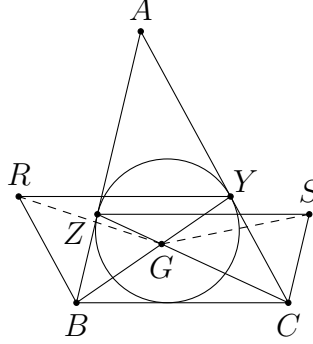
As

$$-\frac{4bc}{(bc-1)(b+c)} \cdot \frac{b+c}{bc+1} + \frac{2bc}{bc-1} = \frac{2bc}{bc+1},$$

the point A does lie on KM' . The proof is complete. $\square$

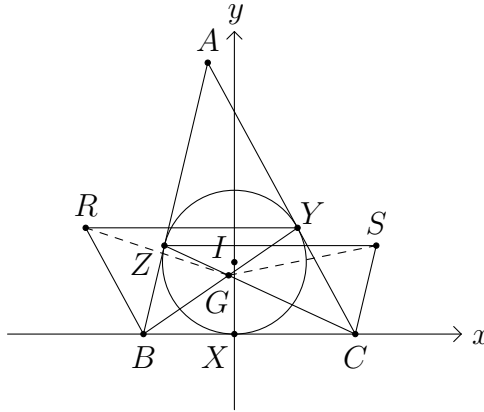
Remarks. As one can see, the major advantage of this setting is that the coordinates of the incentre, the vertices and the contact points of the incircle with the sides have coefficients which are rational functions of two variables. This setting can be used to solve a large class of problems involving the incentre. One may be familiar with the coordinates and equations of some common points and lines after sufficiently many practices.

Problem 5.3. (IMO Shortlist 2009 G3) Let ABC be a triangle. The incircle of ABC touches the sides AB and AC at the points Z and Y , respectively. Let G be the point where the lines BY and CZ meet, and let R and S be points such that the two quadrilaterals $BCYR$ and $BCSZ$ are parallelograms. Prove that $GR = GS$.



Analysis. We see how the same setting as the previous problem can be used to solve this shortlist problem.

Solution.



Let $I(0, 1)$ be the incentre of $\triangle ABC$ and let $B(b, 0)$, $C(c, 0)$. Then

$$m_{AB} = \frac{2m_{BI}}{1 - m_{BI}^2} = -\frac{2b}{b^2 - 1}$$

and hence

$$AB: \quad y = -\frac{2b}{b^2 - 1}x + \frac{2b^2}{b^2 - 1}.$$

Let $X(0, 0)$ be the touching point of the incircle and BC . Note that $XZ \perp BI$, so the equation of XZ is $y = bx$. This easily yields $Z\left(\frac{2b}{b^2 + 1}, \frac{2b^2}{b^2 + 1}\right)$. Then we find

$$S = C + Z - B = \left(c - \frac{b(b^2 - 1)}{b^2 + 1}, \frac{2b^2}{b^2 + 1}\right)$$

and

$$CZ : \quad y = \frac{2b^2}{2b - c(b^2 + 1)}x - \frac{2b^2c}{2b - c(b^2 + 1)}. \quad (1)$$

By symmetry, we find that $R\left(b - \frac{c(c^2 - 1)}{c^2 + 1}, \frac{2c^2}{c^2 + 1}\right)$ and

$$BY : \quad y = \frac{2c^2}{2c - b(c^2 + 1)}x - \frac{2bc^2}{2c - b(c^2 + 1)}. \quad (2)$$

Solving (1) and (2), we have

$$\begin{aligned} x &= \frac{2bc^2(2b - b^2c - c) - 2b^2c(2c - bc^2 - b)}{2c^2(2b - b^2c - c) - 2b^2(2c - bc^2 - b)} \\ &= \frac{2bc(b^2 - c^2)}{2(b^3c^2 - b^2c^3 + b^3 - 2b^2c + 2bc^2 - c^3)} \\ &= \frac{bc(b - c)(b + c)}{(b - c)(b^2c^2 + b^2 - bc + c^2)} \\ &= \frac{bc(b + c)}{b^2c^2 + b^2 - bc + c^2}, \\ y &= \frac{2b^2(x - c)}{2b - c(b^2 + 1)} \\ &= \frac{2b^2}{2b - c(b^2 + 1)} \cdot \left(\frac{bc(b + c)}{b^2c^2 + b^2 - bc + c^2} - c \right) \\ &= \frac{2b^2}{2b - c(b^2 + 1)} \cdot \frac{c(-b^2c^2 + 2bc - c^2)}{b^2c^2 + b^2 - bc + c^2} \\ &= \frac{2b^2c^2}{b^2c^2 + b^2 - bc + c^2}. \end{aligned}$$

This gives $G\left(\frac{bc(b + c)}{b^2c^2 + b^2 - bc + c^2}, \frac{2b^2c^2}{b^2c^2 + b^2 - bc + c^2}\right)$. Next, we find a simpler way to express the condition $GR = GS$. (Here, g_x denotes the x -coordinate of G etc.)

$$\begin{aligned} GR &= GS \\ \Leftrightarrow (g_x - r_x)^2 + (g_y - r_y)^2 &= (g_x - s_x)^2 + (g_y - s_y)^2 \\ \Leftrightarrow (g_x - r_x)^2 - (g_x - s_x)^2 &= (g_y - s_y)^2 - (g_y - r_y)^2 \\ \Leftrightarrow (s_x - r_x)(2g_x - r_x - s_x) &= (r_y - s_y)(2g_y - s_y - r_y) \end{aligned}$$

This provides a factorization of the expression. We just need to evaluate each part separately which is much simpler than expanding everything.

$$\begin{aligned} s_x - r_x &= c - \frac{b(b^2 - 1)}{b^2 + 1} - b + \frac{c(c^2 - 1)}{c^2 + 1} = \frac{2c^3}{c^2 + 1} - \frac{2b^3}{b^2 + 1} \\ &= \frac{2(c^3(b^2 + 1) - b^3(c^2 + 1))}{(b^2 + 1)(c^2 + 1)} = \frac{2(\textcolor{red}{c} - \textcolor{red}{b})(\textcolor{green}{b}^2\textcolor{green}{c}^2 + \textcolor{green}{b}^2 + \textcolor{green}{b}\textcolor{green}{c} + \textcolor{green}{c}^2)}{(\textcolor{brown}{b}^2 + 1)(\textcolor{brown}{c}^2 + 1)} \end{aligned}$$

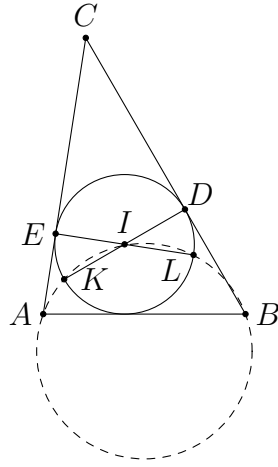
$$\begin{aligned}
2g_x - r_x - s_x &= \frac{2bc(b+c)}{b^2c^2 + b^2 - bc + c^2} - b + \frac{c(c^2-1)}{c^2+1} - c + \frac{b(b^2-1)}{b^2+1} \\
&= \frac{2bc(b+c)}{b^2c^2 + b^2 - bc + c^2} - \frac{2b}{b^2+1} - \frac{2c}{c^2+1} \\
&= \frac{2bc(b+c)}{b^2c^2 + b^2 - bc + c^2} - \frac{2(b+c)(bc+1)}{(b^2+1)(c^2+1)} \\
&= \frac{2(b+c)(bc(b^2+1)(c^2+1) - (bc+1)(b^2c^2 + b^2 - bc + c^2))}{(b^2c^2 + b^2 - bc + c^2)(b^2+1)(c^2+1)} \\
&= -\frac{2(b+c)(b-c)^2}{(b^2c^2 + b^2 - bc + c^2)(b^2+1)(c^2+1)} \\
r_y - s_y &= \frac{2c^2}{c^2+1} - \frac{2b^2}{b^2+1} = \frac{2c^2(b^2+1) - 2b^2(c^2+1)}{(b^2+1)(c^2+1)} = \frac{2(c-b)(c+b)}{(b^2+1)(c^2+1)} \\
2g_y - s_y - r_y &= \frac{4b^2c^2}{b^2c^2 + b^2 - bc + c^2} - \frac{2b^2}{b^2+1} - \frac{2c^2}{c^2+1} \\
&= \frac{4b^2c^2}{b^2c^2 + b^2 - bc + c^2} - \frac{2(2b^2c^2 + b^2 + c^2)}{(b^2+1)(c^2+1)} \\
&= \frac{4b^2c^2(b^2+1)(c^2+1) - 2(2b^2c^2 + b^2 + c^2)(b^2c^2 + b^2 - bc + c^2)}{(b^2c^2 + b^2 - bc + c^2)(b^2+1)(c^2+1)} \\
&= -\frac{2(b^4c^2 - 2b^3c^3 + b^2c^4 + b^4 - b^3c - bc^3 + c^4)}{(b^2c^2 + b^2 - bc + c^2)(b^2+1)(c^2+1)} \\
&= -\frac{2(b-c)^2(b^2c^2 + b^2 + bc + c^2)}{(b^2c^2 + b^2 - bc + c^2)(b^2+1)(c^2+1)}
\end{aligned}$$

It is trivial to check that the products on the two sides are the same, as they have the same factors. $\square$

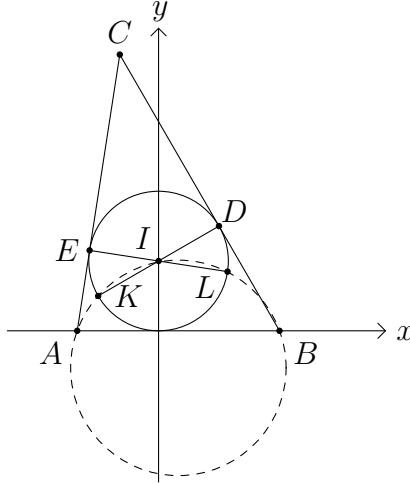
Remarks. Note that $b-c$ and $b+c$ are always factors of some expressions when using coordinate geometry.

Problem 5.4. (IMO Shortlist 2005 G1) Given a triangle ABC satisfying $AC + BC = 3AB$, the incircle of triangle ABC has centre I and touches the sides BC and CA at the points D and E , respectively. Let K and L be the reflections of the points D and E with respect to I . Prove that the points A, B, K, L lie on one circle.

Analysis. Again we shall follow the same setting. Apart from the simple coordinates of the points A, B, C, D, E, I , we emphasize that the side lengths of $\triangle ABC$ also look nice. This is not surprising because the setting ensures that $\tan \frac{A}{2}$ and $\tan \frac{B}{2}$ are rational functions of the variables. This ensures $\sin A, \sin B, \sin C$ are all rational functions, and so are the side lengths by the sine law.



Solution.



Let $I(0, 1)$, $A(a, 0)$, $B(b, 0)$. As in [problem 5.2](#), we find that $C\left(\frac{a+b}{ab+1}, \frac{2ab}{ab+1}\right)$, $D\left(\frac{2b}{b^2+1}, \frac{2b^2}{b^2+1}\right)$ and $E\left(\frac{2a}{a^2+1}, \frac{2a^2}{a^2+1}\right)$. Then we have $K = 2I - D = \left(-\frac{2b}{b^2+1}, \frac{2}{b^2+1}\right)$ and $L\left(-\frac{2a}{a^2+1}, \frac{2}{a^2+1}\right)$.

Now,

$$\begin{aligned}
 & AC + BC = 3AB \\
 \Rightarrow & \sqrt{\left(a - \frac{a+b}{ab+1}\right)^2 + \left(\frac{2ab}{ab+1}\right)^2} + \sqrt{\left(b - \frac{a+b}{ab+1}\right)^2 + \left(\frac{2ab}{ab+1}\right)^2} = 3|b-a| \\
 \Rightarrow & \left|\frac{b}{ab+1}\right| \sqrt{(a^2-1)^2 + (2a)^2} + \left|\frac{a}{ab+1}\right| \sqrt{(b^2-1)^2 + (2b)^2} = 3|b-a| \\
 \Rightarrow & \left|\frac{b}{ab+1}\right| (a^2+1) + \left|\frac{a}{ab+1}\right| (b^2+1) = 3|b-a|.
 \end{aligned}$$

In the setting, we may assume $a < 0 < b$. Also, from the coordinates of C , we have the condition $ab + 1 < 0$ so that its y -coordinate is positive. Therefore,

$$\begin{aligned}
& AC + BC = 3AB \\
\Rightarrow & -\frac{b(a^2 + 1)}{ab + 1} + \frac{a(b^2 + 1)}{ab + 1} = 3(b - a) \\
\Rightarrow & -a^2b - b + ab^2 + a = 3(b - a)(ab + 1) \\
\Rightarrow & (b - a)(ab - 1) = 3(b - a)(ab + 1) \\
\Rightarrow & 2(b - a)(ab + 2) = 0 \\
\Rightarrow & ab = -2
\end{aligned}$$

since $a \neq b$.

We now consider the intersection O of the perpendicular bisectors of AB and KL . Firstly, its x -coordinate is $\frac{a+b}{2}$. Also, the perpendicular bisector of KL is just CI , whose equation is

$$y = \frac{ab - 1}{a + b}x + 1.$$

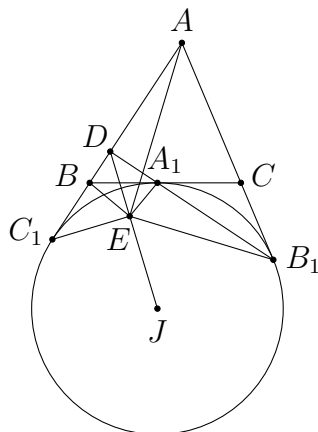
Hence, $O\left(\frac{a+b}{2}, \frac{ab+1}{2}\right)$. It remains to check $OA = OK$.

$$\begin{aligned}
& OA = OK \\
\Leftrightarrow & \left(a - \frac{a+b}{2}\right)^2 + \left(\frac{ab+1}{2}\right)^2 = \left(\frac{a+b}{2} + \frac{2b}{b^2+1}\right)^2 + \left(\frac{ab+1}{2} - \frac{2}{b^2+1}\right)^2 \\
\Leftrightarrow & a^2 - a(a+b) = \frac{2b(a+b)}{b^2+1} + \frac{4b^2}{(b^2+1)^2} - \frac{2(ab+1)}{b^2+1} + \frac{4}{(b^2+1)^2} \\
\Leftrightarrow & -ab(b^2+1) = 2b(a+b) - 2(ab+1) + 4 \\
\Leftrightarrow & -ab(b^2+1) = 2(b^2+1).
\end{aligned}$$

This is true since $ab = -2$. □

Remarks. This solution suggests some conditions on a, b may be added to our setting whenever necessary. In fact, the cases $ab < -1$, $ab = -1$, $ab > -1$ imply respectively C lies above the x -axis (I is the incentre), at infinity (tangents from A and B are parallel, hence is rejected), and below the x -axis (I is the C -excentre). If we do not restrict $a < 0 < b$, we may even obtain cases corresponding to I being the A or B -excentres. In other words, the same construction works equally well for the case of excentres.

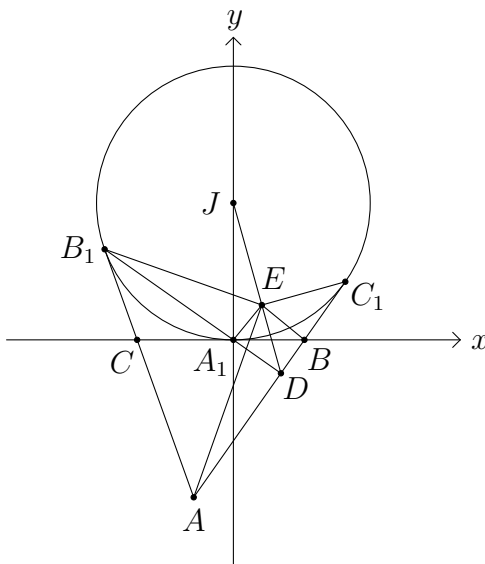
Problem 5.5. (IMO Shortlist 2006 G5) In triangle ABC , let J be the centre of the excircle tangent to side BC at A_1 and to the extensions of sides AC and AB at B_1 and C_1 , respectively. Suppose that the lines A_1B_1 and AB are perpendicular and intersect at D . Let E be the foot of the perpendicular from C_1 to line DJ . Determine the angles $\angle BEA_1$ and $\angle AEB_1$.



Analysis. As explained in the previous problem, the same construction can be applied to excircles. This is mainly because incircle and excircles could hardly be distinguished, especially in coordinate geometry, and they share a lot of common properties. Facts related to the incentre usually yield corresponding results involving excentres. Although we usually draw J ‘under’ BC , there is no need to set its y -coordinate to be negative (say -1).

For this problem, a good drawing could help us guess that both angles are right angles. Anyway, if we cannot guess so and try to apply the angle formula directly, we will note that $\tan \theta$ is undefined, which means the products of the slopes are -1 .

Solution.



Let $A_1(0,0)$, $J(0,1)$, $B(b,0)$, $C(c,0)$. Since we still have $\angle(x, AC) = 2\angle(x, CJ)$ (again x means the x -axis), the same method as in [problem 5.2](#) yields

$$B_1\left(\frac{2c}{c^2+1}, \frac{2c^2}{c^2+1}\right), \quad C_1\left(\frac{2b}{b^2+1}, \frac{2b^2}{b^2+1}\right), \quad A\left(\frac{b+c}{bc+1}, \frac{2bc}{bc+1}\right).$$

From $A_1B_1 \perp AB$, we have $c \cdot \frac{2bc}{(b+c) - b(bc+1)} = -1$ and hence

$$c = \frac{b^2 - 1}{2b}.$$

Solving

$$\begin{aligned} A_1B_1 : \quad y &= cx, \\ AB : \quad y &= -\frac{1}{c}x + \frac{b}{c}, \end{aligned}$$

we get $D = \left(\frac{b}{c^2 + 1}, \frac{bc}{c^2 + 1} \right) = \left(\frac{4b^3}{(b^2 + 1)^2}, \frac{2b^2(b^2 - 1)}{(b^2 + 1)^2} \right)$. Then

$$DJ : \quad y = \frac{b^4 - 4b^2 - 1}{4b^3}x + 1$$

and

$$C_1E : \quad y = -\frac{4b^3}{b^4 - 4b^2 - 1}x + \frac{4b^3}{b^4 - 4b^2 - 1} \cdot \frac{2b}{b^2 + 1} + \frac{2b^2}{b^2 + 1}.$$

After simplification, we get

$$C_1E : \quad y = -\frac{4b^3}{b^4 - 4b^2 - 1}x + \frac{2b^2(b^2 - 1)}{b^4 - 4b^2 - 1}.$$

We can solve the coordinates of E as follows.

$$\begin{aligned} x &= \frac{4b^3(2b^2(b^2 - 1) - (b^4 - 4b^2 - 1))}{(b^4 - 4b^2 - 1)^2 + (4b^3)^2} = \frac{4b^3(b^4 + 2b^2 + 1)}{b^8 + 8b^6 + 14b^4 + 8b^2 + 1} \\ &= \frac{4b^3(b^2 + 1)^2}{(b^4 + 6b^2 + 1)(b^2 + 1)^2} = \frac{4b^3}{b^4 + 6b^2 + 1} \\ y &= \frac{b^4 - 4b^2 - 1}{4b^3} \cdot \frac{4b^3}{b^4 + 6b^2 + 1} + 1 = \frac{2b^2(b^2 + 1)}{b^4 + 6b^2 + 1} \end{aligned}$$

This gives $E \left(\frac{4b^3}{b^4 + 6b^2 + 1}, \frac{2b^2(b^2 + 1)}{b^4 + 6b^2 + 1} \right)$.

We now compute the slopes.

$$\begin{aligned} m_{BE} &= \frac{2b^2(b^2 + 1)}{4b^3 - b(b^4 + 6b^2 + 1)} = -\frac{2b(b^2 + 1)}{b^4 + 2b^2 + 1} = -\frac{2b}{b^2 + 1} \\ m_{EA_1} &= \frac{2b^2(b^2 + 1)}{4b^3} = \frac{b^2 + 1}{2b} = -\frac{1}{m_{BE}} \end{aligned}$$

This gives $\angle BEA_1 = 90^\circ$.

Next, we first write

$$\begin{aligned} A &= \left(\frac{b+c}{bc+1}, \frac{2bc}{bc+1} \right) = \left(\frac{3b^2-1}{b(b^2+1)}, \frac{2(b^2-1)}{b^2+1} \right), \\ B_1 &= \left(\frac{2c}{c^2+1}, \frac{2c^2}{c^2+1} \right) = \left(\frac{4b(b^2-1)}{(b^2+1)^2}, \frac{2(b^2-1)^2}{(b^2+1)^2} \right). \end{aligned}$$

Then we get

$$m_{AE} = \frac{2b(b^2(b^2+1)^2 - (b^2-1)(b^4+6b^2+1))}{4b^4(b^2+1) - (3b^2-1)(b^4+6b^2+1)} = -\frac{2b(3b^4-6b^2-1)}{b^6-13b^4+3b^2+1},$$

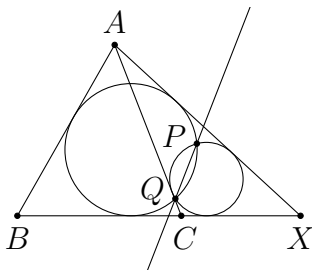
$$m_{EB_1} = \frac{2(b^2(b^2+1)^3 - (b^2-1)^2(b^4+6b^2+1))}{4b(b^2(b^2+1)^2 - (b^2-1)(b^4+6b^2+1))} = \frac{b^6-13b^4+3b^2+1}{2b(3b^4-6b^2-1)} = -\frac{1}{m_{AE}}.$$

This gives $\angle AEB_1 = 90^\circ$. □

Remarks. The calculation could have been simplified if we note that C_1, E, C are collinear (hence E is just the projection of J onto CC_1). This is indeed a general fact even if the condition $AB \perp A_1B_1$ is not satisfied.

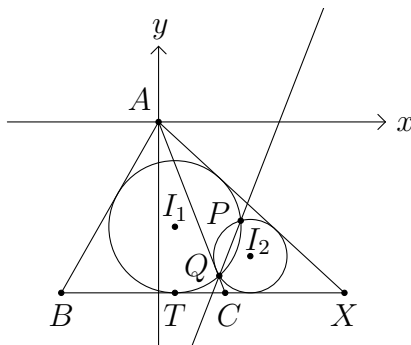
Next, we provide some other methods which can also deal with incentre and excentres.

Problem 5.6. (IMO Shortlist 2004 G7) For a given triangle ABC , let X be a variable point on the line BC such that C lies between B and X and the incircles of the triangles ABX and ACX intersect at two distinct points P and Q . Prove that the line PQ passes through a point independent of X .



Analysis. This time we need to find the incircles of $\triangle ABX$ and $\triangle ACX$. The method used in previous problems cannot be applied as X is a variable point (so that we cannot start from the incentre), and there are two incentres. We shall demonstrate one way to find out the incentre of a fixed triangle using properties of the incentre. Note that the result probably involves radicals.

Solution.



Let $A(0,0)$, $B(b,-1)$, $C(c,-1)$, $X(d,-1)$. (In the following, we need to evaluate many lengths involving A . This is the reason why we put A to be the origin.) Firstly, let I_1, I_2 be the incentres of $\triangle ABX$, $\triangle ACX$ respectively, and let T be the tangent point of $\odot I_1$ to BX .

Observe that $BT = \frac{1}{2}(AB + BX - AX)$. Therefore,

$$T = \left(b + \frac{\sqrt{b^2+1} + (d-b) - \sqrt{d^2+1}}{2}, -1 \right) = \left(\frac{b+d + \sqrt{b^2+1} - \sqrt{d^2+1}}{2}, -1 \right).$$

The inradius of $\triangle ABX$ is equal to $r = \frac{2[ABX]}{AB + BX + AX}$, where $[ABX] = \frac{1}{2}BX$ is the area of $\triangle ABX$. Hence,

$$\begin{aligned} I_1 &= \left(\frac{b+d + \sqrt{b^2+1} - \sqrt{d^2+1}}{2}, -1 + \frac{d-b}{\sqrt{b^2+1} + (d-b) + \sqrt{d^2+1}} \right) \\ &= \left(\frac{b+d + \sqrt{b^2+1} - \sqrt{d^2+1}}{2}, \frac{-\sqrt{b^2+1} - \sqrt{d^2+1}}{d-b + \sqrt{b^2+1} + \sqrt{d^2+1}} \right). \end{aligned}$$

Rationalizing the y -coordinate, this gives

$$\begin{aligned} & \frac{-\sqrt{b^2+1} - \sqrt{d^2+1}}{d-b + \sqrt{b^2+1} + \sqrt{d^2+1}} \cdot \frac{d-b - \sqrt{b^2+1} - \sqrt{d^2+1}}{d-b - \sqrt{b^2+1} - \sqrt{d^2+1}} \\ &= \frac{(b^2+1) + (d^2+1) + (b-d)\sqrt{b^2+1} + (b-d)\sqrt{d^2+1} + 2\sqrt{(b^2+1)(d^2+1)}}{(d-b)^2 - (\sqrt{b^2+1} + \sqrt{d^2+1})^2} \\ &= -\frac{b^2 + d^2 + 2 + (b-d)\sqrt{b^2+1} + (b-d)\sqrt{d^2+1} + 2\sqrt{(b^2+1)(d^2+1)}}{2(bd+1 + \sqrt{(b^2+1)(d^2+1)})} \\ & \quad \times \frac{bd+1 - \sqrt{(b^2+1)(d^2+1)}}{bd+1 + \sqrt{(b^2+1)(d^2+1)}} \\ &= \frac{(b^3d - 2b^2d^2 + bd^3 - b^2 + 2bd - d^2) + (b-d)(bd-d^2)\sqrt{b^2+1} + (b-d)(bd-b^2)\sqrt{d^2+1} - (b^2-2bd+d^2)\sqrt{(b^2+1)(d^2+1)}}{2(b-d)^2} \\ &= \frac{bd-1 + d\sqrt{b^2+1} - b\sqrt{d^2+1} - \sqrt{(b^2+1)(d^2+1)}}{2}. \end{aligned}$$

Then

$$\odot I_1 : \left(x - \frac{b+d + \sqrt{b^2+1} - \sqrt{d^2+1}}{2} \right)^2 + (y - (r-1))^2 = r^2.$$

Note that

$$r = \frac{bd+1 + d\sqrt{b^2+1} - b\sqrt{d^2+1} - \sqrt{(b^2+1)(d^2+1)}}{2}$$

since r is the y -coordinate of I_1 plus 1. We find that

$$\begin{aligned}
& \left(x - \frac{b+d+\sqrt{b^2+1}-\sqrt{d^2+1}}{2} \right)^2 + (y - (r-1))^2 = r^2 \\
\Rightarrow & x^2 + y^2 - (b+d+\sqrt{b^2+1}-\sqrt{d^2+1})x \\
& - (bd-1+d\sqrt{b^2+1}-b\sqrt{d^2+1}-\sqrt{(b^2+1)(d^2+1)})y \\
& + \left(\frac{b+d+\sqrt{b^2+1}-\sqrt{d^2+1}}{2} \right)^2 \\
& - (bd+1+d\sqrt{b^2+1}-b\sqrt{d^2+1}-\sqrt{(b^2+1)(d^2+1)}) + 1 = 0 \\
\Rightarrow & x^2 + y^2 - (b+d+\sqrt{b^2+1}-\sqrt{d^2+1})x \\
& - (bd-1+d\sqrt{b^2+1}-b\sqrt{d^2+1}-\sqrt{(b^2+1)(d^2+1)})y \\
& + \frac{b^2-bd+d^2+1+(b-d)\sqrt{b^2+1}+(b-d)\sqrt{d^2+1}+\sqrt{(b^2+1)(d^2+1)}}{2} = 0. \quad (1)
\end{aligned}$$

Replacing b by c , the equation of $\odot I_2$ is

$$\begin{aligned}
& x^2 + y^2 - (c+d+\sqrt{c^2+1}-\sqrt{d^2+1})x \\
& - (cd-1+d\sqrt{c^2+1}-c\sqrt{d^2+1}-\sqrt{(c^2+1)(d^2+1)})y \\
& + \frac{c^2-cd+d^2+1+(c-d)\sqrt{c^2+1}+(c-d)\sqrt{d^2+1}+\sqrt{(c^2+1)(d^2+1)}}{2} = 0. \quad (2)
\end{aligned}$$

As PQ is the radical axis of the two circles, its equation is obtained by taking the difference of (1) and (2), i.e.

$$\begin{aligned}
& (b-c+\sqrt{b^2+1}-\sqrt{c^2+1})x + (bd-cd+d\sqrt{b^2+1}-d\sqrt{c^2+1}-(b-c)\sqrt{d^2+1} \\
& -\sqrt{(b^2+1)(d^2+1)}+\sqrt{(c^2+1)(d^2+1)})y + \frac{c^2-b^2+bd-cd-(b-d)\sqrt{b^2+1}}{2} \\
& + \frac{(c-d)\sqrt{c^2+1}-(b-c)\sqrt{d^2+1}-\sqrt{(b^2+1)(d^2+1)}+\sqrt{(c^2+1)(d^2+1)}}{2} = 0.
\end{aligned}$$

Due to symmetry, there must be a common factor to all coefficients (say, $b-c$). Therefore, we may try to see if the coefficient of x also divides the other terms. This is indeed the case. After factorization, we get

$$PQ: \quad x + (d-\sqrt{d^2+1})y + \frac{-b-c+2d-\sqrt{b^2+1}-\sqrt{c^2+1}-2\sqrt{d^2+1}}{4} = 0.$$

We have to find a pair of x, y such that this is an identity for any d . One way is to compare the coefficients of $\sqrt{d^2+1}$, which should be zero. That means we need $-y - \frac{1}{2} = 0$. So we take $y = -\frac{1}{2}$. It is then not hard to see that

$$\left(\frac{b+c+\sqrt{b^2+1}+\sqrt{c^2+1}}{4}, -\frac{1}{2} \right)$$

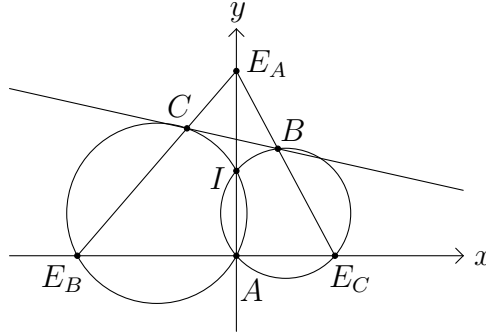
is the fixed point, which depends only on A, B, C . □

Problem 5.7. Let I be the incentre of $\triangle ABC$. Prove that there is a circle tangent to the line BC , the line through A perpendicular to AI , the circumcircle of $\triangle ABI$, and the circumcircle of $\triangle ACI$.

Analysis. The line through A perpendicular to AI is the line joining the B -excentre and C -excentre. Also, the circumcircles of $\triangle ACI$ and $\triangle ABI$ pass through one of these excentres. This suggests we should consider the excentres as well.

For a problem involving the incentre and all the excentres, we may use the fact that I is the orthocentre of $\triangle E_A E_B E_C$. This provides one possible setting of the coordinate system.

Solution.



Let E_A, E_B, E_C be the A, B, C -excentres of $\triangle ABC$ respectively. The line through A perpendicular to AI is $E_B E_C$. Points A, I, B, E_C are concyclic with IE_C being a diameter, and A, I, C, E_B are concyclic with IE_B being a diameter.

Let $A(0,0)$, $E_A(0,1)$, $E_B(b,0)$, $E_C(c,0)$. We may assume $b < 0 < c$ since $\triangle E_A E_B E_C$ is acute. From

$$E_A E_C : y = -\frac{1}{c}x + 1,$$

$$I E_B : y = cx - bc,$$

we get $B\left(\frac{c(bc+1)}{c^2+1}, \frac{c(c-b)}{c^2+1}\right)$ and hence $C\left(\frac{b(bc+1)}{b^2+1}, \frac{b(b-c)}{b^2+1}\right)$.

Also, we easily get $I(0, -bc)$. The centre of (ABI) is the midpoint of IE_C , which has coordinates $\left(\frac{c}{2}, -\frac{bc}{2}\right)$. This gives

$$(ABI) : x^2 + y^2 - cx + bcy = 0. \quad (1)$$

Similarly,

$$(ACI) : x^2 + y^2 - bx + bcy = 0. \quad (2)$$

Also, we have

$$BC: \quad y = \frac{b+c}{1-bc}x - \frac{2bc}{1-bc}. \quad (3)$$

We need to prove that there exists a circle

$$\Gamma: \quad x^2 + y^2 + \alpha x + \beta y + \gamma = 0 \quad (4)$$

which is tangent to (1), (2), (3) and the x -axis. Rewrite (4) as

$$\left(x + \frac{\alpha}{2}\right)^2 + \left(y + \frac{\beta}{2}\right)^2 = \frac{\alpha^2 + \beta^2 - 4\gamma}{4}.$$

This shows the centre of Γ is $\left(-\frac{\alpha}{2}, -\frac{\beta}{2}\right)$ and its radius is $\sqrt{\frac{\alpha^2 + \beta^2 - 4\gamma}{4}}$. We try to look for Γ with centre whose y -coordinate is positive, i.e. $\beta < 0$. Then Γ is tangent to the x -axis iff $-\frac{\beta}{2} = \sqrt{\frac{\alpha^2 + \beta^2 - 4\gamma}{4}}$. As $\beta < 0$, this is equivalent to $\frac{\beta^2}{4} = \frac{\alpha^2 + \beta^2 - 4\gamma}{4}$. This holds iff $\alpha^2 = 4\gamma$.

Next, based on the configuration, it should be possible to find Γ which is internally tangent to (ABI) and externally tangent to (ACI) . Thus, the distance between the centres of Γ and (ABI) is the difference of their radii. Now,

$$\begin{aligned} & \sqrt{\left(\frac{c}{2} + \frac{\alpha}{2}\right)^2 + \left(-\frac{bc}{2} + \frac{\beta}{2}\right)^2} = \left| \frac{c\sqrt{b^2+1}}{2} + \frac{\beta}{2} \right| \\ \Leftrightarrow & \quad (\alpha + c)^2 + (\beta - bc)^2 = (\beta + c\sqrt{b^2+1})^2 \\ \Leftrightarrow & \quad \alpha^2 + 2c\alpha - 2c(b + \sqrt{b^2+1})\beta = 0. \end{aligned} \quad (5)$$

Similarly, we consider

$$\begin{aligned} & \sqrt{\left(\frac{b}{2} + \frac{\alpha}{2}\right)^2 + \left(-\frac{bc}{2} + \frac{\beta}{2}\right)^2} = \frac{-b\sqrt{c^2+1}}{2} - \frac{\beta}{2} \\ \Leftrightarrow & \quad (\alpha + b)^2 + (\beta - bc)^2 = (\beta + b\sqrt{c^2+1})^2 \\ \Leftrightarrow & \quad \alpha^2 + 2b\alpha - 2b(c + \sqrt{c^2+1})\beta = 0. \end{aligned} \quad (6)$$

The difference of (5) and (6) gives $(b-c)\alpha + (c\sqrt{b^2+1} - b\sqrt{c^2+1})\beta = 0$. This means

$$\beta = \frac{b-c}{b\sqrt{c^2+1} - c\sqrt{b^2+1}}\alpha.$$

We substitute this into (5). Note that

$$\begin{aligned} & \alpha^2 + 2c\alpha - 2c(b + \sqrt{b^2+1}) \cdot \frac{b-c}{b\sqrt{c^2+1} - c\sqrt{b^2+1}}\alpha = 0 \\ \Leftrightarrow & \quad \left(\alpha + \frac{2bc(c-b-\sqrt{b^2+1}+\sqrt{c^2+1})}{b\sqrt{c^2+1} - c\sqrt{b^2+1}} \right) \alpha = 0 \\ \Leftrightarrow & \quad \frac{2bc(b-c+\sqrt{b^2+1}-\sqrt{c^2+1})}{b\sqrt{c^2+1} - c\sqrt{b^2+1}} = \alpha. \end{aligned}$$

This suggests we should take

$$\alpha = \frac{2bc(b-c+\sqrt{b^2+1}-\sqrt{c^2+1})}{b\sqrt{c^2+1}-c\sqrt{b^2+1}}, \quad \beta = \frac{2bc(b-c)(b-c+\sqrt{b^2+1}-\sqrt{c^2+1})}{(b\sqrt{c^2+1}-c\sqrt{b^2+1})^2}.$$

From above, it is obvious that (5) and (6) are satisfied. That means Γ is tangent to (ABI) and (ACI) . Also, we can define γ correspondingly using $\alpha^2 = 4\gamma$. It now remains to check it is also tangential to BC . Consider the projection of the centre $\left(-\frac{\alpha}{2}, -\frac{\beta}{2}\right)$ of Γ onto BC . The equation of the perpendicular line from this centre to BC is

$$y = -\frac{1-bc}{b+c}x - \frac{(1-bc)\alpha}{2(b+c)} - \frac{\beta}{2}.$$

Solving this and (3), we see that the intersection point P is given by

$$\left(\frac{4bc(b+c) - (1-bc)^2\alpha - (b+c)(1-bc)\beta}{2((b+c)^2 + (1-bc)^2)}, -\frac{4bc(1-bc) + (b+c)(1-bc)\alpha + (b+c)^2\beta}{2((b+c)^2 + (1-bc)^2)} \right).$$

We need to check that the distance from P to the centre of Γ is equal to the radius of Γ , i.e. $\left(x + \frac{\alpha}{2}\right)^2 + \left(y + \frac{\beta}{2}\right)^2 = \left(-\frac{\beta}{2}\right)^2$. This is equivalent to

$$\left(\frac{(b+c)(4bc + (b+c)\alpha - (1-bc)\beta)}{2((b+c)^2 + (1-bc)^2)} \right)^2 + \left(\frac{(1-bc)(-4bc - (b+c)\alpha + (1-bc)\beta)}{2((b+c)^2 + (1-bc)^2)} \right)^2 = \frac{\beta^2}{4}.$$

This holds iff

$$((b+c)^2 + (1-bc)^2)(4bc + (b+c)\alpha - (1-bc)\beta)^2 = ((b+c)^2 + (1-bc)^2)^2\beta^2,$$

i.e.

$$(4bc + (b+c)\alpha - (1-bc)\beta)^2 = (b^2+1)(c^2+1)\beta^2.$$

Indeed, we find

$$\begin{aligned} \frac{4bc + (b+c)\alpha - (1-bc)\beta}{\beta} &= \frac{4bc}{\beta} + (b+c) \cdot \frac{\alpha}{\beta} - (1-bc) \\ &= \frac{2(b\sqrt{c^2+1} - c\sqrt{b^2+1})^2}{(b-c)(b-c+\sqrt{b^2+1}-\sqrt{c^2+1})} + \frac{(b+c)(b\sqrt{c^2+1} - c\sqrt{b^2+1})}{b-c} - (1-bc) \\ &= \frac{(b\sqrt{c^2+1} - c\sqrt{b^2+1})(b^2 - c^2 + (b-c)\sqrt{b^2+1} + (b-c)\sqrt{c^2+1})}{(b-c)(b-c+\sqrt{b^2+1}-\sqrt{c^2+1})} - (1-bc) \\ &= \frac{(b\sqrt{c^2+1} - c\sqrt{b^2+1})(b+c+\sqrt{b^2+1}+\sqrt{c^2+1})}{b-c+\sqrt{b^2+1}-\sqrt{c^2+1}} - (1-bc) \\ &= \frac{-(c^2+1)\sqrt{b^2+1} + (b^2+1)\sqrt{c^2+1} + (b-c)\sqrt{(b^2+1)(c^2+1)}}{b-c+\sqrt{b^2+1}-\sqrt{c^2+1}} \\ &= \sqrt{(b^2+1)(c^2+1)} \end{aligned}$$

as desired. This completes the proof. $\square$

Remarks. There could be at most four possible choices of the desired circle depending on the internally/externally tangent conditions.

In view of this construction, we may say that problems about incentre and excentres are actually equivalent to problems about orthocentre, and coordinate geometry works equally well for both kinds of problems!

Links

Problems

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- [1.4](#): Euler line
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- [2.1](#): Symmetry
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- 5.7: Incentre and excentres

Terminologies and Notations

- $\ell_1 \cap \ell_2$
- Cartesian plane
- coordinate
- coordinate geometry
- coordinate plane
- general form
- origin
- quadrant
- slope
- slope-intercept form
- x -axis
- x -coordinate
- y -axis
- y -coordinate
- y -intercept

「夫難平者，事也；凡事如是，難可逆見。」

《三國演義》 / 《後出師表》